Two-Phase Certificate Bounds for Backtracking Frank–Wolfe under Function-Dependent Curvature

Extended technical report

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Abstract

We consider convex minimization over a compact convex set with a linear minimization oracle and curvature bounded by $\|\nabla^2 f(x)\| \le H_0 + H_1(f(x) - f_Q^*)$. The optimal value is unavailable to the algorithm. We use accumulated linear lower bounds to construct a computable upper certificate. Standard quadratic backtracking attains certificate accuracy ε in $O(H_0 D^2/\varepsilon + (1 + H_1 D^2)\log(U_0/\varepsilon))$ linear-oracle calls, up to an explicit initialization term and the final verification call, for $0 < \varepsilon < U_0$. It requires no curvature or optimal-value input. A phase argument bounds the cost of estimates that lag behind decreasing local curvature. All function trials cost at most $1+2N$ plus an explicit logarithmic overhead after N updates. The same linear-oracle order holds for two explicit steps, exact segment minimization, and a certificate-scaled backtracking variant; normalization is not needed for this order. For standard away-step backtracking, we extend the certificate bound by counting atom removals. Under a geometric error bound, a second phase estimate gives logarithmic accuracy dependence and separates the H_1 warm-up term from the H_0 logarithmic term, without parameter inputs. We also give a corrected lower certificate for deterministically bounded oracle errors. A local gap envelope removes strong convexity from the primal-to-certificate conversion; strong convexity remains one sufficient condition for the geometric error bound. Experiments cover seven families, including real-data Poisson regression with positive H_1. Controlled initialization, radius, and objective-unit comparisons separate oracle budgets and reveal important baseline sensitivities. A Poisson activity sweep excludes repeated starting vertices from its primary summaries. Two further count datasets show that online certificate attainment can lag the independently bounded primal error. Causal block-averaged tangent controls do not yield earlier stopping on these paths; a standard known-strong-convexity control resolves much of the lag on one dataset, using additional information. No uniform empirical superiority of certificate initialization or priority over all adaptive analyses is established. Independent proof checking and a complete priority audit remain necessary.

1 Introduction and scope

Frank–Wolfe methods replace projections by linear minimization over the feasible region. The resulting oracle model is attractive when linear minimization is inexpensive and sparse convex combinations are useful; see the survey [10]. A global smoothness constant can nevertheless make a short step conservative. Function-dependent curvature offers a different way to choose an explicit step.

The condition studied here is motivated by the loss-suboptimality curvature model of Alimisis, Islamov, and Lucchi [6]. Related developments include generalized-smooth clipped methods [9] and accelerated full-gradient and coordinate methods [8]. Vyguzov and Stonyakin [7] analyze Frank–Wolfe with curvature depending on the gradient norm, including adaptive parameter selection.

Other practical choices include backtracking [11], secant line search [13], and the recent auto-conditioned framework [16]. Analytic steps derived from other curvature models also exist, for example generalized self-concordance [14]. Adaptive Bregman steps treat relative smoothness [20], while adaptive conditional gradient sliding reuses gradients across inner linear-oracle iterations [21]. Their assumptions and oracle units require a separate comparison; the present study does not establish superiority over these broader frameworks.

Extended-curvature error-bound rates [2] and restart schemes adaptive to error bounds [3] further constrain any priority claim. Linear objective- and FW-gap convergence beyond global gradient smoothness also precedes this work for log-homogeneous barriers [4]. Appendix A records the precise comparisons used here.

The candidate contribution is an analysis of standard FW and away-step backtracking under (1), with separate dependence on H_0, H_1, a computable stopping criterion, an initialization term, and a bound on all function trials. It is a result about an existing method; it does not require a new step-size rule. The linear certificate, scalar differential comparison, halving argument, quadratic acceptance test, and telescoping count of failures are standard tools. The quadratic step has the same guarantee as the hyperbolic step, as does classical exact line search in the linear-oracle model. An elementary function in a step rule is therefore not the main claim. We do not establish optimal oracle complexity, an accelerated rate for all smooth functions, or priority over all adaptive analyses. The related-work search in this working version is targeted and incomplete.

2 Setting and computable certificates

Let $Q \subset \mathbb{R}^d$ be nonempty, compact, and convex, of Euclidean diameter D. Let f be convex and twice continuously differentiable on a neighborhood of Q, and set $f_Q^* = \min_{x\in Q} f(x)$. Assume $H_0, H_1 \geq 0$ satisfy

$$\left\|\nabla^2 f(z)\right\|_2 \leq H_0 + H_1\big(f(z) - f_Q^*\big), \qquad z \in Q. \tag{1}$$

Only the explicit steps require these constants as inputs; backtracking does not. This is a condition relative to the *constrained* minimum. A condition relative to a smaller unconstrained optimum cannot be substituted with the same H_0 automatically.

At $x_k \in Q$, compute

$$s_k \in \arg\min_{v\in Q} \langle \nabla f(x_k), v\rangle, \quad d_k = s_k - x_k, \quad G_k = -\langle \nabla f(x_k), d_k\rangle. \tag{2}$$

Convexity gives $f(x_k) - G_k \leq f_Q^*$ and $0 \leq f(x_k) - f_Q^* \leq G_k$. Consequently

$$\ell_k = \max_{0\leq j\leq k}\{f(x_j) - G_j\}, \qquad U_k = f(x_k) - \ell_k \tag{3}$$

satisfy $f(x_k) - f_Q^* \leq U_k \leq G_k$. A valid known lower bound may be included in the maximum, but is not required. The algorithm stops at $U_k \leq \varepsilon$. When $G_k = 0$, convexity already certifies optimality.

3 Two explicit steps and a local progress bound

We suppress iteration indices. Write $q = \|d\|$, and let $f(x) - f_Q^* \leq V \leq G$. In the algorithm one may take $V = U$ or $V = G$.

Lemma 1 (Hyperbolic upper model)**.** *If $H_1 > 0$ and $c = q\sqrt{H_1} > 0$, then, for $0 \leq t \leq 1$,*

$$f(x + td) - f(x) \leq M_V(t) = \left(V + \frac{H_0}{H_1}\right)(\cosh(ct) - 1) - \frac{G}{c}\sinh(ct). \tag{4}$$

For $H_1 = 0$ the upper model is $-Gt + H_0 q^2 t^2/2$.

Proof. Let $y(t) = f(x + td) - f_Q^*$ and $\delta = y(0)$. Assumption (1) implies $y'' \le q^2 H_0 + c^2 y$, while $y'(0) = -G$. The equality solution with these initial data is

$$u_\delta(t) = \left(\delta + \frac{H_0}{H_1}\right)\cosh(ct) - \frac{G}{c}\sinh(ct) - \frac{H_0}{H_1}.$$

For $w = y - u_\delta$, variation of constants expresses $w(t)$ as the integral of $w'' - c^2 w \le 0$ against the nonnegative kernel $\sinh(c(t-r))/c$. Thus $w \le 0$. Subtract the initial value and use $\delta \le V$ and $\cosh(ct) - 1 \ge 0$. □

Let $B = V + H_0/H_1$ and $r = G/(cB)$. When these quantities are positive, the model minimizer on $[0, 1]$ is

$$\alpha_H = \begin{cases} 1, & r \ge 1, \\ \min\{1, \operatorname{artanh}(r)/c\}, & 0 < r < 1. \end{cases} \tag{5}$$

The quadratic step is

$$a = (H_0 + H_1 V)q^2, \qquad \alpha_Q = \min\{1, G/a\}. \tag{6}$$

For $H_1 = 0$ the two rules coincide. When $a = 0$, use a full step. Zero gap or zero diameter is handled before evaluating these expressions.

Lemma 2 (Local progress without an exponential factor)**.** *For either step, if $a > 0$,*

$$f(x) - f(x + \alpha d) \ge \frac{G}{2}\min\{1, G/a\}. \tag{7}$$

If $a = 0$, a full step decreases the function by at least G.

Proof. For the hyperbolic model, $M''' = c^2 M'$, $M'(0) = -G$, and $M''(0) = a$. Its derivative is nonpositive up to its selected minimizer α_H. Also $\alpha_Q \le \alpha_H$: this follows from $\operatorname{artanh}(r) \ge r$ for $r < 1$, and from $\alpha_H = 1$ for $r \ge 1$. Hence $M'' \le a$ on $[0, \alpha_H]$. Therefore

$$M(\alpha_H) \le M(\alpha_Q) \le -G\alpha_Q + \tfrac{1}{2}a\alpha_Q^2 \le -\tfrac{1}{2}G\alpha_Q.$$

Lemma 1 proves the assertion for α_H.

For α_Q, set $\psi(t) = f(x+td) - f(x)$. While $\psi \le 0$, (1) gives $\psi'' \le a$. Initially $\psi'(0) = -G < 0$. A first return to zero at $0 < t < \min\{1, 2G/a\}$ would imply $\psi(t) \le -Gt + at^2/2 < 0$, a contradiction. Thus this quadratic bound holds up to $\alpha_Q \le G/a$, proving (7). For $H_1 = 0$ the bound is the ordinary quadratic calculation. If $a = 0$ and $G > 0$, a valid V cannot be zero, since $V = 0$ would imply optimality and hence $G = 0$. Consequently $a = 0$ forces $H_0 = H_1 = 0$, so f is affine on the segment and the full step is valid. □

The proof only bounds curvature on a descent segment. Bounding $\cosh(ct) - 1$ uniformly on the entire segment would instead introduce an unnecessary factor $e^{D\sqrt{H_1}}$. The quadratic model is not asserted for arbitrary $t \in [0, 1]$.

4 Complexity of a computable stopping certificate

The algorithm computes (2)–(3), stops if $U_k \le \varepsilon$, chooses $V_k \in [U_k, G_k]$, and updates $x_{k+1} = x_k + \alpha_k d_k$ with either rule. In particular, retaining the lower-bound history is sufficient; no explicit restarting is needed.

Theorem 3. *Under the stated assumptions, both rules decrease f, and U_k is nonincreasing. If $D > 0$ and $0 < \varepsilon < U_0$, define $J = \lceil \log_2(U_0/\varepsilon) \rceil$. The algorithm reaches $U_N \le \varepsilon$ within*

$$N \le \left\lceil (5 + 4H_1 D^2)J + \frac{16 H_0 D^2}{\varepsilon} \right\rceil \tag{8}$$

updates. Each explicit-step implementation uses $N + 1$ function, gradient, and linear-oracle evaluations including the initial point and final check. If $U_0 \le \varepsilon$, no update is needed; if $D = 0$ the problem has one feasible point. If $H_0 = H_1 = 0$, one full step suffices.

Proof. Monotonicity follows from Lemma 2; ℓ_k is nondecreasing, so $U_k = f(x_k) - \ell_k$ is nonincreasing. Since $V_k \le G_k$, $q_k \le D$, and $G_k \ge U_k$, local progress yields

$$f_k - f_{k+1} \ge \frac{1}{2} \min\left\{ G_k, \frac{G_k^2}{D^2(H_0 + H_1 G_k)} \right\} \ge \frac{1}{2} \min\left\{ U_k, \frac{U_k^2}{D^2(H_0 + H_1 U_k)} \right\}. \tag{9}$$

The second inequality uses monotonicity of $z^2/(H_0 + H_1 z)$ for $z > 0$. The affine case has already been handled.

Consider a phase starting with $U \le b$ and ending at $U \le b/2$. While it continues, each update reduces f by at least

$$p(b) = \min\left\{ \frac{b}{4}, \frac{b^2}{8D^2 H_0 + 4D^2 H_1 b} \right\}.$$

Total decrease from the start of the phase cannot exceed $f_{\text{start}} - f_Q^* \le U_{\text{start}} \le b$. Thus the phase ends within $\lfloor b/p(b) \rfloor + 1 \le 5 + 4D^2 H_1 + 8D^2 H_0/b$ updates. Apply this to $b_j = U_0/2^j$, $j = 0, \ldots, J-1$. Skipped phases cost zero. Finally, $\sum_{j=0}^{J-1} 1/b_j = (2^J - 1)/U_0 < 2/\varepsilon$, which proves (8). □

For $H_0 = 0$, the bound is logarithmic. This is a restricted curvature class, not a logarithmic guarantee for arbitrary globally smooth objectives. For example $f(x) = t + t^3$, $t = 1/2 - x$, on $[-1/2, 1/2]$ satisfies $|f''| = 6t \le 6f$, and has $H_0 = 0, H_1 = 6$. Our numerical checks include this limit. For general $H_0 > 0$, the inverse-accuracy term remains.

Proposition 4 (Scope of the preceding bound)**.** *Classical Frank–Wolfe with exact minimization over each oracle segment and the accumulated certificate* (3) *satisfies Theorem 3's update bound without knowing H_0, H_1.*

Proof. The exact minimizer performs at least as well as the feasible quadratic step with $V = U_k$. Its local decrease thus satisfies (9), and the phase proof applies unchanged. □

This observation follows from our argument; it is not a bibliographic priority claim. It also gives no bound on the function evaluations needed to obtain an exact scalar minimizer. Finite-tolerance golden-section search in the experiment is not its literal exact-arithmetic implementation.

5 Adaptation without curvature parameters

Assume a positive known upper bound D on the diameter. Maintain the same accumulated certificate, and initialize a dimensionless estimate $\theta_{-1} = 1$. Whenever $U_k > \varepsilon$, start with $\theta = \theta_{k-1}/2$, set

$$L = \theta U_k / D^2, \qquad \alpha = \min\{1, G_k/(Lq_k^2)\},$$

and test

$$f(x_k + \alpha d_k) \le f(x_k) - \alpha G_k + \tfrac{1}{2} L \alpha^2 q_k^2. \tag{10}$$

Double θ on rejection, and store the accepted value as θ_k. Accepted function values are reused. This is a standard quadratic acceptance test [11] with a certificate-dependent normalization; adaptation to unknown smoothness is not new by itself [16].

Lemma 5 (Accepted-curvature envelope)**.** *The search terminates in exact arithmetic and its accepted curvature obeys*

$$L_k \le \max\left\{ \frac{U_k}{2^{k+1} D^2},\ 2(H_0 + H_1 U_k) \right\}. \tag{11}$$

Proof. For $L \ge H_0 + H_1 U_k$, the initial-descent argument of Lemma 2 implies (10). Hence doubling terminates. The sufficient threshold for θ is $T_k = D^2(H_0/U_k + H_1)$. Acceptance gives monotone f and therefore nonincreasing U_k; thus T_k is nondecreasing. Inductively suppose $\theta_{k-1} \le \max\{2^{-k}, 2T_{k-1}\}$. If the first trial is accepted, $\theta_k = \theta_{k-1}/2 \le \max\{2^{-k-1}, T_k\}$. Otherwise the last rejected value is smaller than T_k, so the accepted value is smaller than $2T_k$. The initial case starts from $\theta_{-1} = 1$. Multiply the resulting bound by U_k/D^2 to obtain (11). □

Theorem 6 (Unknown curvature and total trial cost)**.** *Under the assumptions of Theorem 3, with $J = \lceil \log_2(U_0/\varepsilon) \rceil$, certificate-scaled backtracking attains $U_N \le \varepsilon$ within*

$$N \le \left\lceil (7 + 8H_1 D^2) J + 32 H_0 D^2/\varepsilon \right\rceil \tag{12}$$

updates. It uses $N + 1$ gradients and linear-oracle calls. For the actual number $N \ge 1$ of updates, its function calls, including all rejected trials, satisfy

$$N_f \le 1 + 2N + \left\lceil \log_2 \max\{1, 2D^2(H_0/\varepsilon + H_1)\} \right\rceil. \tag{13}$$

For $N = 0$, only the initial function value is needed. The statements use exact oracles and exact arithmetic. The H_0, H_1 in the bound describe the objective; they are not inputs to this adaptive algorithm.

Proof. Acceptance guarantees a decrease of at least $G_k \alpha_k/2$. From (11), for $k \ge 0$,

$$L_k \le 2H_0 + (2H_1 + 1/(2D^2)) U_k.$$

Thus the progress and phase argument of Theorem 3 applies with effective constants $2H_0$ and $2H_1 + 1/(2D^2)$, proving (12). These effective constants only bound accepted step curvatures; they are not a new Hessian assumption.

Let r_k count rejected trials. The recursion is exactly $\theta_k = 2^{r_k - 1}\theta_{k-1}$. Telescoping gives

$$\sum_{k=0}^{N-1} (r_k + 1) = 2N + \log_2 \theta_{N-1}.$$

The last source certificate exceeds ε. Hence Lemma 5 bounds the last normalized estimate by $\max\{1, 2D^2(H_0/\varepsilon + H_1)\}$. Add the initial function value and reuse each accepted value to obtain (13). □

The recurrence is invariant under $\widetilde{f} = af + b$, $a > 0$, with tolerance scaled by a: G, U, L scale together and α, θ do not. As with other sufficient-decrease tests, floating-point differences near an optimum can violate exact-arithmetic comparisons. In a planted active example, certificate scales near 10^{-9} can correspond to decreases below 10^{-18}. Envelope tests stop at 10^{-6}; the main study uses 10^{-5}. Roundoff is not included in the theorem.

5.1 The same order for ordinary backtracking

Certificate normalization is not necessary for the two-phase order. Consider the standard quadratic backtracking scheme of [11], specialized to initial trial $L_{k-1}/2$ and doubling on rejection, with the same test (10) and certificate (3). Let $L_{-1} = M > 0$ be arbitrary. The algorithm needs neither curvature constants nor a diameter input; D enters only its analysis. The optional initialization $M = U_0/D^2$ uses a diameter bound and also makes the trajectory invariant under positive objective rescaling, with the tolerance scaled accordingly.

Theorem 7 (Ordinary backtracking)**.** *Under the assumptions of Theorem 3, let* $J = \lceil \log_2(U_0/\varepsilon) \rceil$ *and* $P = \lceil \log_2 \max\{1, MD^2/U_0\} \rceil$*. Ordinary backtracking reaches* $U_N \le \varepsilon$ *within*

$$N \le P + \left\lceil (6 + 16H_1D^2)J + 32H_0D^2/\varepsilon \right\rceil \tag{14}$$

updates, using $N + 1$ *gradients and linear-oracle calls. For the actual* $N \ge 1$*, all function calls satisfy*

$$N_f \le 1 + 2N + \lceil \log_2 \max\{1, 2(H_0 + H_1U_0)/M\} \rceil . \tag{15}$$

For $M = U_0/D^2$*,* $P = 0$ *and the logarithmic overhead in* (15) *is independent of* ε*.*

Proof. As before, every $L \ge H_0 + H_1U_k$ passes the test. If a step has at least one rejection, its accepted curvature is strictly less than $2(H_0 + H_1U_k)$. If there is no rejection, $L_k = L_{k-1}/2$. Acceptance implies monotone f and U.

Let $b_j = U_0/2^j$, $j = 0, \dots, J-1$, and assign a source iterate to phase j when $b_j/2 < U_k \le b_j$. Sources with $U_k > \varepsilon$ belong to these phases. Define the positive caps

$$C_j = \max\{2(H_0 + H_1b_j),\ b_j/(2D^2)\}.$$

They satisfy $C_j/2 \le C_{j+1} \le C_j$. Call a step a lag step if its accepted $L_k > C_j$ in its source phase. Such a step has no rejection, so its curvature halves. Once $L_k \le C_j$, all subsequent steps in the same phase also satisfy this bound: their initial trial is at most $C_j/2$, and a trial accepted after rejection is below $2(H_0 + H_1U_k) \le C_j$.

For completeness, before a step in phase j give the previous estimate the nonnegative integer potential

$$p = \max\{0, \lceil \log_2(L_{k-1}/C_j) \rceil\}.$$

A lag step reduces this potential by one; a non-lag step leaves potential zero before any phase change. A jump over m phases can increase it by at most m, because the cap falls by at most 2^m. Initially $p \le P + 1$, since $C_0 \ge U_0/(2D^2)$. Source phase indices increase by at most $J - 1$ in total. Thus at most $P + J$ lag steps occur, also on any prefix of the run.

On a non-lag step in phase j, acceptance, $G_k \ge U_k > b_j/2$, and $q_k \le D$ give a decrease of at least

$$p_j = \min\{b_j/4,\ b_j^2/(8C_jD^2)\}.$$

All steps decrease f, and the total decrease during this phase is at most b_j. Its non-lag steps therefore number at most

$$\lfloor b_j/p_j \rfloor + 1 \le 5 + 16H_1D^2 + 16H_0D^2/b_j.$$

Sum over phases and use $\sum_j 1/b_j < 2/\varepsilon$, then add the lag bound. The same argument bounds every prefix of an unfinished run, establishing termination and (14).

Finally the accepted estimates obey

$$L_k \le \max\{M/2^{k+1},\ 2(H_0 + H_1U_0)\}.$$

Writing r_k for the number of rejections gives $L_k = 2^{r_k-1}L_{k-1}$ and hence exactly $\sum_{k<N}(r_k+1) = 2N + \log_2(L_{N-1}/M)$. The preceding envelope proves (15) after adding the initial value. Accepted values are cached. □

This is an analysis of an existing algorithm, not a new backtracking rule. The constants above are conservative and do not imply a practical ranking of the two variants. A complete literature audit is still needed to determine whether this specialization or an equivalent argument has already appeared. In particular, the telescoping trial-count identity is standard in backtracking analyses [11].

Proposition 8 (Why an initialization term remains)**.** *Let $Q = [-1, 1]$, $f(x) = -cx$ with $c > 0$, $x_0 = -1$, and $L_{-1} = M > 0$. Although $H_0 = H_1 = 0$, ordinary halving backtracking needs exactly*

$$N = \left\lceil \log_2 \left(1 + \frac{M}{c} \left(1 - \frac{\varepsilon}{2c} \right) \right) \right\rceil$$

updates for $0 < \varepsilon < 2c$. In particular, halving U_0 can require $\Theta(\log(M/c))$ updates as $M/c \to \infty$.

Proof. Every quadratic test accepts its first trial, since f is affine. Thus $L_k = M/2^{k+1}$ and $x_{k+1} = \min\{1, x_k + c/L_k\}$. The linear lower bound is always the exact optimum $-c$, so

$$x_N = \min\{1, -1 + (2c/M)(2^N - 1)\}, \qquad U_N = c(1 - x_N).$$

Solve $U_N \le \varepsilon$ for the integer N. This concerns the initialization penalty of this particular backtracking rule; it is not a lower bound for arbitrary first-order or linear-oracle algorithms. With $M = U_0/D^2 = c/2$, the first step is full and solves the problem. □

6 Away steps: certificates and removal of active atoms

Let $Q = \mathrm{conv}(\mathcal{A})$ for a finite atom set. Maintain $x = \sum_{a\in S} \lambda_a a$ with positive weights summing to one. Besides the FW atom s, choose $v \in \arg\max_{a\in S} \langle \nabla f(x), a\rangle$. Use the larger descent gap of $s - x$ and $x - v$; ties choose FW. Denote the chosen direction by d and its gap by $g = -\langle \nabla f(x), d\rangle$. Its feasible cap is $\alpha_{\max} = 1$ for FW and $\lambda_v/(1 - \lambda_v)$ for an away step. A singleton active set takes a FW step. The stopping certificate always uses the *full FW gap*, not the selected direction's gap. In particular $g \ge G \ge U$.

This is classical away-step FW with standard quadratic backtracking [5, 11]. Start from $L_{-1} = M > 0$; each update halves the previous estimate, doubles on rejection, and uses

$$\alpha = \min\{\alpha_{\max}, g/(L\|d\|^2)\}$$

in (10), with G there replaced by g. An away update at its cap is a *drop step*; all other updates are called good. Searching the active set incurs additional work, separate from one full LMO per iterate. The bounds in this section use exact arithmetic and exact full and active-set linear oracles.

Lemma 9 (Local acceptance and counting drops)**.** *The quadratic test passes for $L \geq H_0 + H_1\delta$, where $\delta = f(x) - f_Q^*$. This also holds when $\alpha_{\max} > 1$. If m good updates occur, the total number of updates is at most $2m + |S_0| - 1$.*

Proof. On the feasible segment $[0, \alpha_{\max}]$, the same initial-descent argument as in Lemma 2 applies with the chosen g. More explicitly, while the function is at most $f(x)$ its directional second derivative is bounded by $(H_0 + H_1\delta)\|d\|^2$. The resulting quadratic upper bound is strictly below $f(x)$ for $0 < t \leq \min\{\alpha_{\max}, g/(L\|d\|^2)\}$, preventing a first return to the initial level. Thus the bound holds through the trial. Every drop removes an atom, whereas a FW step adds at most one and is good. Active-set cardinality is positive, so drops number at most the number of FW steps plus $|S_0|-1$. □

Theorem 10 (Convex certificate bound with away steps)**.** *With J, P as in Theorem 7, away-step backtracking reaches $U_N \leq \varepsilon$ in at most*

$$2\left[P + \left\lceil (6 + 16H_1D^2)J + 32H_0D^2/\varepsilon \right\rceil\right] + |S_0| - 1$$

updates. It uses $N + 1$ gradients and full LMOs, and, for actual $N \geq 1$,

$$N_f \leq 1 + 2N + \lceil \log_2 \max\{1, 2(H_0 + H_1U_0)/M\} \rceil .$$

Proof. Use the same phase caps and integer potential as in Theorem 7, counting lag steps across *all* updates. On a good away step the quadratic minimizer is not clipped; on a full FW step acceptance decreases f by at least $g/2$. Hence a good update with $L \leq C_j$ decreases f by at least $\min\{b_j/4, b_j^2/(8C_jD^2)\}$. Count only these good updates using the phase budget; drops also decrease f and cannot increase that budget. The resulting bound on good updates is the ordinary bound. Apply Lemma 9. The accepted-curvature envelope and exact telescoping trial identity hold across drops as well, proving the function-call bound. □

The known-H AFW quadratic rule, with $L = H_0 + H_1U_k$ and the same cap, similarly inherits Theorem 3's bound on good updates. Its total updates are at most twice that bound plus $|S_0| - 1$, and it uses $N + 1$ function calls. The stronger theorem below concerns ordinary backtracking, whose accepted estimates can follow the true local error.

6.1 A logarithmic certificate bound under a geometric error bound

Assume the selected gaps satisfy the standard geometric error bound

$$g_k^2 \geq 2\kappa\big(f(x_k) - f_Q^*\big), \qquad \kappa > 0. \tag{16}$$

Strong convexity with parameter $\mu > 0$ is one sufficient condition: for a finite polytope, one may use $\kappa = \mu W^2/4$, where W is its pyramidal width. The pairwise gap squared is at least $2\mu W^2\delta$ and the selected gap is at least half that pairwise gap [5, 11]. No numerical width estimate fitted to a trajectory is used here. Strong convexity is not required separately when (16) is available; geometric error-bound rates are established results [2, 3].

Proposition 11 (A standard quadratic-growth sufficient condition)**.** *Let $X^* = \arg\min_Q f$ and suppose that, on the initial sublevel set,*

$$f(x) - f_Q^* \geq \frac{\gamma}{2}\operatorname{dist}(x, X^*)^2, \qquad \gamma > 0.$$

For a polytope of pyramidal width W, condition (16) *holds along monotone iterates with $\kappa = \gamma W^2/16$. No unique minimizer or positive strong-convexity parameter is required.*

Proof. Let p be the pairwise gap for the current active set, and take a nearest $x^* \in X^*$ when $\delta > 0$. The standard polytope scaling inequality and convexity give

$$p \geq W \frac{\langle \nabla f(x), x - x^* \rangle}{\|x - x^*\|} \geq W \frac{\delta}{\operatorname{dist}(x, X^*)} \geq W\sqrt{\gamma\delta/2}.$$

The selected FW/away gap is at least $p/2$. Squaring proves $g^2 \geq \gamma W^2 \delta/8 = 2\kappa\delta$; zero error is immediate. Monotone iterates stay in the sublevel set where growth is assumed. □

This is the known scaling/Holderian-error-bound implication at exponent 1/2 [3], Lemmas 3.4 and 3.6, with constants translated to our selected-gap convention. It is not a new geometric result. The direct strong-convexity implication above has a better constant. For example, $f(x) = \cosh(x_1) - 1$ on a two-dimensional ℓ_1 ball satisfies quadratic growth with $\gamma = 1$ and a non-singleton optimal set, while its Hessian has a zero eigenvalue.

Lemma 12 (From primal error to the stopping certificate)**.** *Under* (1)*, every feasible point with primal error $\delta = f(x) - f_Q^*$ satisfies*

$$U \leq G \leq \widehat{B}(\delta) := \begin{cases} \sqrt{2C(\delta)\delta}, & 2\delta \leq C(\delta), \\ \delta + C(\delta)/2, & 2\delta > C(\delta), \end{cases} \qquad C(\delta) = D^2(H_0 + H_1\delta).$$

This is at most the earlier conservative envelope $B(\delta) = \max\{2\delta, \sqrt{2C(\delta)\delta}\}$. For $D > 0$, put $a = D^2H_0$, $b = D^2H_1$. Then $\delta \leq \eta_\varepsilon$ implies $U \leq \varepsilon$, where

$$\eta_\varepsilon = \begin{cases} (\varepsilon - a/2)/(1 + b/2), & b < 2 \text{ and } \varepsilon(2 - b) \geq 2a, \\ \varepsilon^2/(a + \sqrt{a^2 + 2b\varepsilon^2}), & \text{otherwise.} \end{cases}$$

In particular $H_0 = H_1 = 0$ gives $\eta_\varepsilon = \varepsilon$. Neither strong convexity nor uniqueness of the minimizer is needed.

Proof. Use the local quadratic model on the hypothetical step $t = \min\{1, G/C(\delta)\}$, with a full step if $C(\delta) = 0$. The initial-descent argument in Lemma 2 proves that the model is valid through this step; $\|s - x\| \leq D$. Feasibility and the exact model decrease give

$$\delta \geq \Phi_C(G) := \begin{cases} G^2/(2C), & G \leq C,\ C > 0, \\ G - C/2, & G > C \text{ or } C = 0. \end{cases}$$

Inverting this increasing function of G gives $\widehat{B}$. If $2\delta \leq C$, the square-root branch is unchanged; otherwise $\delta + C/2 < 2\delta$ and exceeds $\sqrt{2C\delta}$. Solving $\widehat{B}(\eta) = \varepsilon$ gives the stated threshold. The full-step root applies exactly when $b < 2$ and $\varepsilon(2 - b) \geq 2a$; otherwise rationalization gives the quadratic root. This threshold is at least the earlier conservative $\min\{\varepsilon/2, \varepsilon^2/(a + \sqrt{a^2 + 2b\varepsilon^2})\}$, with the zero-curvature convention for that expression. □

The two-branch inversion already appears in [1, Eq. (15)]. Here the feasible-descent argument uses a source-dependent local curvature bound and retains the exact full-step decrease. For $H_0 = 0 < H_1$, the threshold is $\varepsilon/(1 + b/2)$ when $b < 2$, and $\varepsilon/\sqrt{2b}$ when $b \geq 2$. For fixed $H_0 > 0$ and small ε, it is still of order ε^2.

The square-root branch cannot in general be replaced by a linear primal-to-gap bound. For $f(x) = x^2/2$ on $[-1, 1]$, at $x = t > 0$,

$$\delta = t^2/2, \qquad G = t(1 + t), \qquad G/\delta = 2(1 + t)/t \longrightarrow \infty.$$

At the first observation the scalar certificate also equals G. This example has strong convexity one. It limits pointwise conversion for this certificate, not the iteration complexity of a method or the benefit of an accumulated history. In fact, $\widehat{B}$ is attained on both branches when $H_1 = 0$: for $Q = [-D/2, D/2]$, a source at $D/2$ and a quadratic of curvature H_0 centred inside Q give the square-root branch; centres to the left of $-D/2$ give the full-step branch. Thus this refinement is sharp for the quadratic descent inequality; no optimal-envelope claim is made for general $H_1 > 0$.

Theorem 13 (Phase bound under a geometric error bound)**.** *Assume additionally* $H_0 + H_1 > 0$ *and* (16)*. For* $0 < \varepsilon < U_0$*, set*

$$J_* = \lceil \log_2(U_0/\eta_\varepsilon) \rceil , \qquad P_* = \lceil \log_2 \max\{1, M/[2(H_0 + H_1 U_0)]\} \rceil .$$

The same algorithm reaches $U_N \le \varepsilon$ *within*

$$N \le 2 \left\lceil P_* + (6 + 4H_0 D^2/\kappa) J_* + 8H_1 D^2 U_0/\kappa \right\rceil + |S_0| - 1.$$

The preceding function-call bound remains valid. The algorithm needs none of H_0, H_1, κ, f_Q^* *as inputs. For fixed problem and initialization, the accuracy dependence is logarithmic.*

Proof. Analyze phases of the *true* error, with $b_j = U_0/2^j$, stopping at η_ε. These phases are for analysis only. Use positive caps $C_j = 2(H_0 + H_1 b_j)$, which decrease by at most a factor two. The initial potential is at most P_* in virtual phase zero. Allowing initially skipped phases, its total increases number at most $J_* - 1$; at most $P_* + J_*$ lag steps occur.

On a controlled good step, acceptance and (16) give

$$f_k - f_{k+1} \ge \delta_k \min\{1/2, \kappa/(C_j D^2)\} > \min\{b_j/4, \kappa b_j/(2C_j D^2)\}.$$

The phase budget b_j permits at most $5 + 4H_0 D^2/\kappa + 4H_1 D^2 b_j/\kappa$ such steps. Sum with $\sum_j b_j < 2U_0$, add lag steps, and apply the drop count. Lemma 12 ensures the actual stopping certificate has passed by that point. No stronger assertion is made for general convex functions: the extra geometric error bound matters. □

The geometric inequality, drop counting, and linear convergence of adaptive AFW are established results. The candidate extension here is the separated curvature dependence and its conversion to this particular certificate, not a new away-step method or a priority claim. The requirement $H_0 > 0$ in the earlier working version was stronger than needed: the proof only needs positive caps at positive source errors, which $H_1 > 0$ also ensures. For example, $f(x) = t + t^3$, $t = 1/2 - x$, on $[-1/2, 1/2]$ has $H_0 = 0, H_1 = 6$, zero strong-convexity modulus, and a positive selected-gap geometric constant. The all-affine case $H_0 = H_1 = 0$ is covered by the convex backtracking bound, without invoking these caps. For comparison, substituting $L_Q = H_0 + H_1 \max_Q(f - f_Q^*)$ in a global-smoothness linear rate leaves the entire global curvature multiplying the accuracy logarithm. Here the explicit H_1 phase cost is proportional to U_0, with further dependence inside J_* through the certificate bridge. The new bridge uses no μ or initial-level curvature $H_0 + H_1 U_0$; those quantities are unnecessary for converting primal error into this stopping criterion. This comparison does not rule out deriving an equivalent specialization from a prior adaptive AFW proof; its priority remains unresolved.

7 Causal certificate controls on an unchanged path

We describe two standard lower-bound constructions used as stopping controls. Neither is a new primal–dual method. Write $g_j = \nabla f(x_j)$ and $a_j = f(x_j) - \langle g_j, x_j \rangle$. For a completed block B of K

consecutive source observations, set

$$\bar{a}_B = K^{-1} \sum_{j \in B} a_j, \qquad \bar{g}_B = K^{-1} \sum_{j \in B} g_j, \qquad b_B = \bar{a}_B + \min_{v \in Q} \langle \bar{g}_B, v \rangle .$$

The last minimization costs one additional full LMO. Disjoint blocks start at source $j = 0$; an incomplete block is not queried.

Proposition 14 (Elementary stopping control)**.** *Let $f(x_k)$ be nonincreasing along a fixed feasible path, and let ℓ_k be the accumulated scalar tangent lower bound. Define*

$$\widetilde{\ell}_k = \max\{\ell_k,\ b_B : B \text{ completed by } k\}, \qquad \widetilde{U}_k = f(x_k) - \widetilde{\ell}_k.$$

Then $\delta_k \le \widetilde{U}_k \le U_k$, and $\widetilde{U}_k$ is nonincreasing. Its first crossing of a prescribed tolerance occurs no later on this path than that of U_k. Up to source k, exactly $\lfloor (k+1)/K \rfloor$ extra LMOs are used, with no extra function or gradient calls.

Proof. Convexity gives $f(v) \ge a_j + \langle g_j, v \rangle$ for each source and every $v \in Q$. Averaging and minimizing gives $b_B \le f_Q^*$. Taking the maximum preserves validity and is nondecreasing. Subtracting it from the nonincreasing function values proves the assertions; the oracle count follows from the number of completed blocks. □

In particular, if the original algorithm stops after N updates, this control uses at most $N + 1 + \lfloor (N+1)/K \rfloor$ total LMOs and no more function/gradient calls than the original run. This is a pathwise comparison, not an improved worst-case convergence rate. Earlier stopping need not reduce the total LMO count. The monitor affects neither the line search nor the initial curvature estimate.

Average tangent bounds need not dominate the best scalar tangent bound. For $f(x) = x^2/2$ on $[-1, 1]$, the two-source histories $(-1, 1)$ and $(-1, -1/2)$ give block bounds $-1/2$ and $-17/16$, versus scalar maxima $-3/2$ and $-5/8$, respectively. The maximum in Proposition 14 prevents deterioration. For a globally defined differentiable convex f, Fenchel equality and Jensen give

$$b_B = -K^{-1} \sum_{j \in B} f^*(g_j) - \sigma_Q(-\bar{g}_B) \ \le \ -f^*(\bar{g}_B) - \sigma_Q(-\bar{g}_B) = \psi(\bar{g}_B).$$

Here f^* is the convex conjugate, distinct from the scalar optimum f_Q^*. Pedregosa et al. [11, Theorem 3] already analyze an averaged-gradient Fenchel dual for adaptive FW variants, with their specified weights. Our uniform, disjoint-block tangent computation does not evaluate that dual or inherit its theorem.

A separate control receives a valid strong-convexity constant $\mu > 0$. Minimizing the quadratic lower model over the ambient space gives

$$b_j^\mu = f(x_j) - \frac{\|g_j\|^2}{2\mu} \ \le f_Q^*, \qquad \widetilde{\ell}_k^\mu = \max\{\ell_k, \max_{j \le k} b_j^\mu\}.$$

This standard bound is recorded, for example, in [12, Appendix B]; it remains valid for a constrained minimum because minimizing the model over $\mathbb{R}^d$ only lowers it. It requires no extra LMO but uses more information than the block controls. At a boundary solution a nonzero gradient can keep it weak. All assertions here use exact arithmetic; the experiment audits floating-point bounds against independent reference intervals.

8 A corrected certificate with inexact information

The convergence theorem assumes exact oracles. The following separate statement only establishes validity of a lower certificate.

Proposition 15. *Suppose* $\|\widehat{g} - \nabla f(x)\| \le \rho$, $\langle \widehat{g}, s\rangle \le \min_{v\in Q}\langle \widehat{g}, v\rangle + \eta$, *and* $|\widehat{f} - f(x)| \le \nu$, *with known deterministic bounds. Then*

$$\ell^{\mathrm{safe}} = \widehat{f} + \langle \widehat{g}, s - x\rangle - \rho D - \eta - \nu \le f_Q^*.$$

Proof. For $v \in Q$, convexity and Cauchy–Schwarz imply

$$f(v) \ge \widehat{f} - \nu + \langle \widehat{g}, v - x\rangle - \rho D \ge \widehat{f} - \nu + \langle \widehat{g}, s - x\rangle - \eta - \rho D.$$

Minimize over v. A maximum of such lower bounds is still valid. □

An upper certificate at a noisy function evaluation is $\widehat{f} + \nu - \max_j \ell_j^{\mathrm{safe}}$. A variance estimate is not a deterministic norm bound. No noisy-direction complexity theorem is claimed here. Without the correction, even $f(x) = x^2/2$ at $x = 1$, with $\widehat{g} = 0$ on $[-1, 1]$, gives a false lower bound $1/2 > f_Q^* = 0$.

9 Experiments

9.1 Problems with analytically verified assumptions

On $Q = \{x : \|x\|_1 \le R\}$, the oracle returns a signed coordinate vertex at a largest-magnitude gradient coordinate. Let

$$\phi(x) = \frac{1}{m}\sum_{i=1}^{m}[\cosh(a_i^\top(x - x^*)) - 1] + \frac{\mu}{2}\|x - x^*\|^2,$$

where $\mu > 0$ and $x^* \in Q$. Valid constants are

$$H_0 = \mu + \lambda_{\max}(A^\top A/m), \qquad H_1 = \max_i \|a_i\|^2.$$

Indeed, decompose the Hessian into its value at x^* and the positive semidefinite sum weighted by $\cosh(a_i^\top(x - x^*)) - 1$. The latter has norm at most $H_1\phi(x)$.

An optimum placed on the boundary can still have zero gradient. To test genuinely active constraints, choose $\|x^*\|_1 = R$, $v \in \partial\|x^*\|_1$, and $\lambda > 0$, and set

$$f(x) = \phi(x) - \lambda v^\top(x - x^*). \tag{17}$$

On Q, $v^\top(x - x^*) \le 0$. Hence $f \ge \phi \ge 0$, $f_Q^* = 0$, and x^* is its unique constrained minimizer. Its gradient is $-\lambda v \ne 0$. The KKT multiplier is positive, and x^* also minimizes $f(x) + \lambda\|x\|_1$ over $\mathbb{R}^d$. The Hessian is unchanged, so the same constants satisfy (1) on Q. These are planted examples, not naturally occurring application instances.

The original interior/boundary cases use $d = 12$, $m = 10$, $R = 1.5$, $\mu = 0.08$, and independent $a_{ij} \sim N(0, 0.22^2)$. The start is Re_1. The optimum has entries $(0.3, -0.25, 0.2)$ on coordinates $(2, 6, 10)$ in the interior case, and $(0.6, -0.5, 0.4)$ otherwise. Active cases use $\lambda = 0.15$, signs on the support of x^*, and independent $v_j \sim \mathrm{Uniform}(-0.6, 0.6)$ off that support. The scaled case

changes the Gaussian standard deviation to 0.9; the high-dimensional case uses $d = 60, m = 40$ and standard deviation 0.22.

As a real-data control, we solve ℓ_1-constrained logistic regression on the Wisconsin Diagnostic Breast Cancer dataset [15], using the copy bundled with scikit-learn. For each seed a stratified 80% training partition is standardized using training statistics only. The objective is mean binary logistic loss plus $0.01 \|x\|^2 /2$, on a ball of radius 2. A valid bound is $H_0 = L = 0.25\lambda_{\max}(A^\top A/m) + 0.01, H_1 = 0$. Thus this is a standard-smoothness control, not evidence of gains for positive H_1. SLSQP on positive/negative split variables supplies an independent feasible reference and its linear lower certificate; the largest reference interval width in the ten runs is $3.1 \cdot 10^{-9}$. These reference quantities are used only for post-run error reporting.

9.2 Protocol and findings

We use ten matched seeds, eleven methods, a cap of 1000 updates, and a common stopping condition $U \le 10^{-5}$: 660 runs in total. The three hyperbolic variants use respectively $V = G$, accumulated certificates, and an additional valid lower bound 0. The extra information in the last variant is explicitly disclosed. The quadratic variant uses accumulated certificates. Baselines include global-L short steps, $2/(k+2)$, Armijo with initial step 1, shrinkage 1/2 and parameter 10^{-4}, adaptive quadratic backtracking, and Algorithm 2 of [7] with $\rho = 3$ and both initial parameters equal to 1. The scalar adaptive-L estimate starts at 1, is halved on each update, and doubled on rejection.

The secant implementation is a safeguarded variant inspired by [13], not a reproduction of the authors' code. It checks the endpoint, starts at 0 and 10^{-3}, uses a derivative-root tolerance $10^{-10} \max\{1, G\}$, and maintains a bracket with bisection when necessary, including every eighth trial. Golden-section search checks both endpoints and uses an absolute interval tolerance 10^{-10}. The script records all internal oracle calls, reuses accepted values, and logs line-search safeguards. No baseline hyperparameter is tuned on the reported tasks.

Table 1 reports selected scenarios and methods. Complete results, including the additional lower-bound variant and remaining scenarios, are in the accompanying CSV files. The hyperbolic and quadratic methods are nearly indistinguishable on these instances. In the interior case the accumulated-certificate hyperbolic method reaches the certificate tolerance in a median 136.5 updates, versus 170 for global L and 27 for Armijo. On the active scaled family, no hyperbolic-memory run certifies the target within the cap: median error is $3.64 \cdot 10^{-3}$, versus $5.18 \cdot 10^{-2}$ for global L and $3.03 \cdot 10^{-6}$ for the diminishing step. The latter certifies only four of ten runs, illustrating the distinction between small primal error and an available stopping certificate. An exact numerical line minimizer need not yield the best finite-budget multi-step trajectory.

9.3 Reproducibility and limitations

From the PART3 directory, after installing its pinned requirements, run

```
python Work/benchmark.py --seeds 10 --max-iter 1000 --tolerance 1e-5
```

The recorded environment uses Python 3.13.5, NumPy 2.5.2, SciPy 1.16.3, Matplotlib 3.10.9, and scikit-learn 1.8.0, with BLAS/OpenMP thread environment variables set to 1. The manifest records platform details, parameters, reference intervals, and source SHA-256 hashes. Timings are one run per instance, include logging, and exclude problem/reference construction. They are exploratory; inter-seed variation does not replace repeated timing on each instance. Function, gradient, and LMO costs are reported separately, without assigning arbitrary relative weights.

Table 1: Recorded comparison. Each cell is the median over ten seeds; successes count runs with $U \leq 10^{-5}$ within 1000 updates. Capped runs remain in all medians. Error is a reference-based upper bound on suboptimality. Function and gradient calls include line-search trials.

Method	Successes	Updates	U	Error $\leq$	f calls	∇f calls
interior						
Hyperbolic, memory	10/10	136.5	9.48e-06	1.81e-09	137.5	137.5
Quadratic, memory	10/10	137.0	9.41e-06	1.72e-09	138.0	138.0
Global L	10/10	170.0	9.33e-06	1.96e-09	171.0	171.0
$2/(k+2)$	1/10	1000.0	4.02e-05	1.57e-07	1001.0	1001.0
Armijo	10/10	27.0	7.10e-06	1.42e-10	220.0	28.0
Adaptive L	10/10	85.0	9.34e-06	1.31e-09	168.5	86.0
Adaptive (L_0, L_1)	10/10	71.5	8.37e-06	1.04e-09	263.0	72.5
Safeguarded secant	10/10	46.0	8.56e-06	8.79e-10	47.0	158.5
Golden-section search	10/10	46.0	8.56e-06	8.79e-10	2347.0	47.0
active						
Hyperbolic, memory	0/10	1000.0	3.64e-04	2.71e-04	1001.0	1001.0
Quadratic, memory	0/10	1000.0	3.64e-04	2.71e-04	1001.0	1001.0
Global L	0/10	1000.0	4.59e-04	3.71e-04	1001.0	1001.0
$2/(k+2)$	5/10	612.5	2.25e-05	1.91e-08	613.5	613.5
Armijo	10/10	19.0	6.55e-06	2.02e-10	141.0	20.0
Adaptive L	0/10	1000.0	2.23e-04	1.63e-04	1998.5	1001.0
Adaptive (L_0, L_1)	0/10	1000.0	1.95e-04	1.54e-04	3601.0	1001.0
Safeguarded secant	1/10	1000.0	1.63e-04	9.51e-05	1001.0	3188.5
Golden-section search	1/10	1000.0	1.63e-04	9.51e-05	51001.0	1001.0
active_scaled						
Hyperbolic, memory	0/10	1000.0	4.25e-03	3.64e-03	1001.0	1001.0
Quadratic, memory	0/10	1000.0	4.25e-03	3.64e-03	1001.0	1001.0
Global L	0/10	1000.0	6.62e-02	5.18e-02	1001.0	1001.0
$2/(k+2)$	4/10	1000.0	1.91e-04	3.03e-06	1001.0	1001.0
Armijo	1/10	1000.0	4.78e-04	2.50e-04	9758.0	1001.0
Adaptive L	0/10	1000.0	1.77e-03	1.32e-03	2002.0	1001.0
Adaptive (L_0, L_1)	0/10	1000.0	1.30e-03	1.03e-03	3604.5	1001.0
Safeguarded secant	0/10	1000.0	1.25e-03	7.05e-04	1001.0	3920.0
Golden-section search	0/10	1000.0	1.25e-03	7.05e-04	51001.0	1001.0
breast_cancer						
Hyperbolic, memory	0/10	1000.0	8.67e-03	8.41e-03	1001.0	1001.0
Quadratic, memory	0/10	1000.0	8.67e-03	8.41e-03	1001.0	1001.0
Global L	0/10	1000.0	8.67e-03	8.41e-03	1001.0	1001.0
$2/(k+2)$	0/10	1000.0	2.80e-05	5.60e-07	1001.0	1001.0
Armijo	8/10	66.0	8.63e-06	1.40e-08	610.0	67.0
Adaptive L	0/10	1000.0	1.55e-04	1.30e-04	1997.0	1001.0
Adaptive (L_0, L_1)	0/10	1000.0	1.04e-04	9.12e-05	3597.5	1001.0
Safeguarded secant	0/10	1000.0	7.28e-05	5.23e-05	1001.0	4087.0
Golden-section search	0/10	1000.0	7.28e-05	5.23e-05	51001.0	1001.0

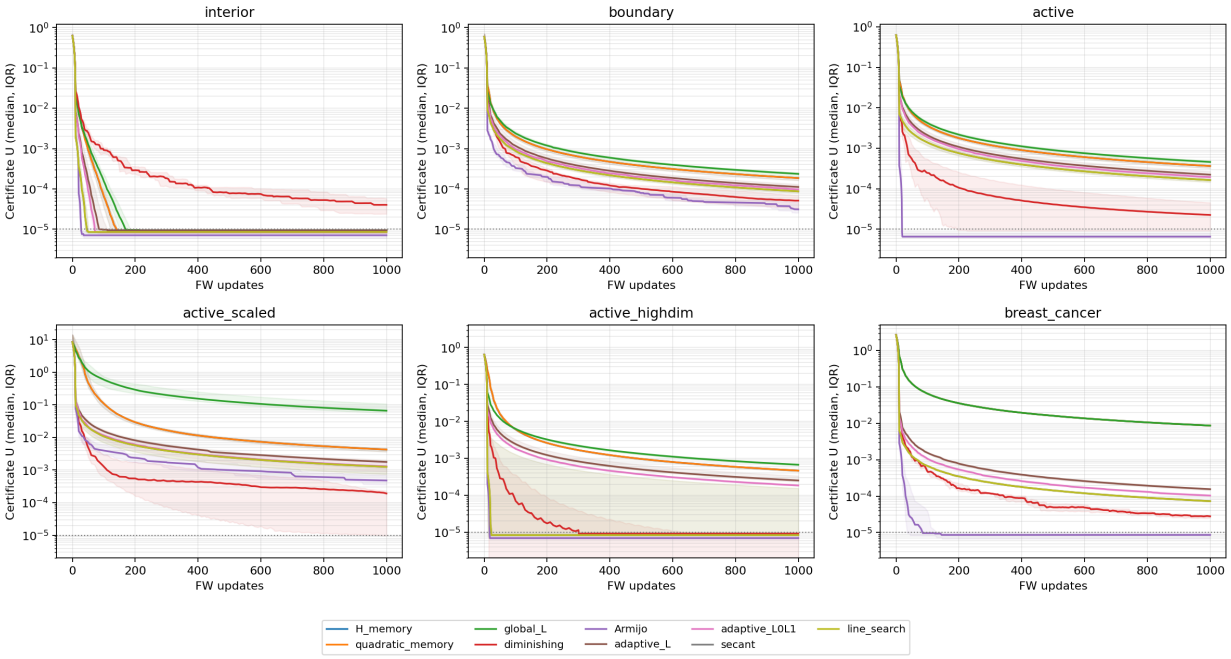


Figure 1: Median certificate and interquartile band across ten seeds. Completed runs are carried forward at their final value. The figure uses trace samples every ten updates plus the final point, linearly interpolated for plotting only. The vertical display range is clipped below $2 \cdot 10^{-6}$; actual values, including zero certificates, are retained in the CSV files. The horizontal reference is the stopping tolerance 10^{-5}.

Fifty-eight automated tests compare the analytic minimizer with numerical scalar minimization and check numerical derivatives, KKT conditions, curvature bounds, local progress, certificates, oracle counts, objective-shift invariance, independence from reference values, the $H_1 \to 0$ limit, $H_0 = 0$, and deterministic error corrections. Additional checks cover the adaptive envelope, exact telescoping identity, total function-call bound, objective rescaling, unknown-curvature inputs, and a positive-H_1 real-data problem, plus matching first steps in the initialization ablation. Boundary tests cover both attainable quadratic gap-envelope branches and the geometric theorem with $H_0 = 0$. A separate exact-rational audit checks 8370 finite configurations, including 57989 abstract curvature-state visits, phase-budget constants, gap inversion and atom-removal counts. Its archived source and hashes are replayed by the validator. The abstract state model dominates possible curvature histories but does not assert realizability of every state by an objective. Neither finite numerical checks nor this rational audit establish a universal theorem or replace independent proof review.

Generative-tool disclosure. An OpenAI Codex assistant was used substantially for literature triage, proof development, implementation, experiment analysis, manuscript drafting and verification automation. Explicit derivations, archived sources and executable checks are provided for scrutiny; this internal process is not independent mathematical peer review. Responsibility for submitted scientific claims rests with the named authors.

9.4 Additional study: adaptation and real count data

A separate recorded study repeats the controls with the adaptive method and adds Poisson regression: 12 methods, seven scenarios, ten seeds, or 840 runs. These include reruns of previous controls, not 840 independent new problem instances. The previous 660-run experiment and exact source snapshots are retained. New results are in `Work/adaptive_results/`.

For the official UCI Bike Sharing daily data [17], use the first 584 of 731 chronological observations. Training-only standardized features are temperature, apparent temperature, humidity, windspeed, working-day, holiday and year indicators, plus an intercept. The target components `casual` and `registered` are excluded. Let $t_i = y_i/\bar{y}$, where y_i is the count and $\bar{y}$ its training mean. On an ℓ_1 ball of radius $R = 1.5$, minimize

$$f(x) = \frac{1}{m}\sum_i \left[e^{a_i^\top x} - t_i a_i^\top x + t_i \log t_i - t_i\right] + \frac{\mu}{2}\|x\|^2, \quad \mu = 0.01.$$

This is a Poisson-deviance objective with offset $\log \bar{y}$, divided by $\bar{y}$, plus the quadratic penalty. Seeds vary only the starting signed vertex of this fixed training objective. No forecasting performance claim or test-set evaluation is made.

For $H_1 = \max_i \|a_i\|^2$ and $C = m^{-1}\sum_i (t_i \log t_i - t_i)$, $\left\|\nabla^2 f\right\| \le \mu + H_1 m^{-1}\sum_i e^{a_i^\top x}$. Also $f_Q^* \le f(0) = 1 + C$ and $m^{-1}\sum_i t_i a_i^\top x \le R m^{-1}\sum_i t_i \|a_i\|_\infty$. Thus (1) holds with the explicit, conservative constant

$$H_0 = \mu + H_1\left(1 + \frac{R}{m}\sum_i t_i \|a_i\|_\infty\right).$$

Here $H_0 \approx 165.37$, $H_1 \approx 49.39$, while the guaranteed global L is approximately 8098.06. The independent SLSQP reference is polished by Newton steps when interior and is accompanied by its FW-gap interval. The reference is unavailable to every FW solver.

Table 2 gives selected results. On the real count problem, certificate-scaled backtracking and the chosen adaptive-L baseline each certify eight of ten starts. Their respective median function-call counts are 860 and 1789, including capped runs. Armijo certifies all ten starts, using 4399

median function calls and 310 updates. This is a promising comparison with one backtracking configuration, not uniform superiority or a paired speedup for every start. On the scaled active family, the new method certifies no run within the cap. All 70 runs of the new method obey the total trial bound in (13); this is a trajectory check, not a proof by experiment. The real-data study covers only one dataset and radius with positive H_1. The next subsection adds auto-conditioned FW [16].

Table 2: Additional 840-run study: selected comparisons. Medians over ten seeds include capped runs; tolerance 10^{-5} and cap 1000 updates. On Bike Sharing, seeds vary the starting vertex of one fixed training objective.

Method	Successes	Updates	U	f calls	∇f calls
interior					
Hyperbolic, known H_0, H_1	10/10	136.5	9.48e-06	137.5	137.5
Adaptive L	10/10	85.0	9.34e-06	168.5	86.0
Certificate-scaled backtracking	10/10	75.0	8.71e-06	168.0	76.0
Armijo	10/10	27.0	7.10e-06	220.0	28.0
active_scaled					
Hyperbolic, known H_0, H_1	0/10	1000.0	4.25e-03	1001.0	1001.0
Adaptive L	0/10	1000.0	1.77e-03	2002.0	1001.0
Certificate-scaled backtracking	0/10	1000.0	1.47e-03	2014.0	1001.0
Armijo	1/10	1000.0	4.78e-04	9758.0	1001.0
bike_poisson					
Hyperbolic, known H_0, H_1	0/10	1000.0	1.04e-01	1001.0	1001.0
Adaptive L	8/10	893.5	9.85e-06	1789.0	894.5
Certificate-scaled backtracking	8/10	419.0	9.41e-06	860.0	420.0
Armijo	10/10	310.0	9.93e-06	4399.0	311.0

A 210-run ablation repeats the same seven families with three methods. The new control uses the same initial curvature U_0/D^2 and the same first trial as the proposed method, followed by ordinary halving of L without current-certificate scaling. On Bike Sharing its median function count is 1002.5, with six certified starts, versus 860 and eight for the proposed method. The original baseline's count of 1789 therefore partly reflects initialization. The median of the ten paired function-call ratios is 0.953, with a descriptive 95% percentile-bootstrap interval $[0.535, 2.303]$ (10000 resamples, seed 7103, capped runs included). This wide interval includes 1; stable empirical superiority is not established. On the interior family the corresponding median ratio is 1.050 with interval $[0.824, 1.184]$. These are descriptive intervals without multiple-comparison correction. The ablation reuses the main instances and is not independent evidence from new datasets. Theorem 7 now proves the same two-phase order for ordinary backtracking, including an arbitrary positive initial estimate.

9.5 Ordinary backtracking and an auto-conditioned comparator

A separate 350-run extension uses the same seven families, ten seeds, tolerance, and cap. It compares ordinary backtracking initialized at 1 or U_0/D^2, certificate scaling, Armijo, and AC-FW [16]. Of these runs, 280 repeat existing deterministic trajectories with additional logging; only the 70 AC-FW runs add a new method on these same instances. They are not new independent datasets.

AC-FW is our independent implementation of Algorithms 1 and 2 in [16], using its equation

(6) with a fixed $\delta = 1$:

$$r_t = 1 - \frac{1}{(t+1)\log^2(t+3)}.$$

We do not claim to reproduce the authors' numerical configuration. Its compulsory move from x_{-1} to an LMO vertex x_0, both function and gradient evaluations, and the initial LMO are charged. Available linear lower bounds from both points enter the certificate history. After initialization, each outer iteration makes one function trial; strict decrease accepts the point, and rejection keeps the old point and cached gradient. Rejections consume the iteration cap. Consequently $N_f = N + 2$, $N_{\mathrm{LMO}} = N + 2$, and $N_g = 2 + N_{\mathrm{accepted}}$, including the final certificate check. The initial move changes the starting trajectory; its effect is included in the algorithm comparison, not isolated in an initialization ablation. Our cosh and Poisson losses are not globally gradient-Lipschitz on $\mathbb{R}^d$, unlike the assumption in the cited paper; their use here is an empirical comparison on the compact feasible set, not a claim that its global theorem applies verbatim.

Table 3 shows strong competing results. On the active family AC-FW certifies all ten runs, with medians 18.5 function and gradient calls; certificate scaling certifies one run within the cap. On the scaled active family neither adaptive variant nor AC-FW certifies any run, and AC-FW's median certificate is worse. On Bike Sharing AC-FW uses 639 median function calls and certifies seven starts, versus 860 calls and eight starts for certificate scaling; its median gradient count is higher (639 versus 420). Capped runs remain in every median, so these figures must be read together with successes and both oracle budgets. Only five outer trials are rejected across the 70 AC-FW runs. This study gives no uniform practical ranking or evidence that the normalization is necessary.

We replay all 140 ordinary-backtracking runs from their archived source and check every accepted curvature, decrease, phase classification, lag budget, telescoping trial identity, and trial bound. The replay matches recorded objectives, certificates, and all three oracle counts exactly. All these runs have zero lag steps relative to the conservative analytic phase caps; they do not test sharpness of the lag bound or the large worst-case iteration constants. A separate affine example tests more than twenty lag steps and the exact formula in Proposition 8. Five hundred abstract trajectories stress the potential argument with abrupt phase jumps and varying initial scales. These checks support implementation consistency, not independent mathematical validation or a priority claim.

9.6 Separating vertex initialization from the update rule

The preceding AC-FW comparison changes both initialization and the subsequent rule. We therefore run a controlled 2×2 study on the same 70 instances: the original point or the compulsory LMO vertex, combined with AC updates or ordinary backtracking (BT). All four cells pay for two function values, two gradients, and two LMOs at the two preprocessing points. They retain both scalar lower bounds and the same secant curvature. BT's stored initial estimate is twice the AC estimate, so its first halving gives exactly the same first trial. Afterwards BT halves and doubles its estimate, while AC uses the fixed damping above. AC/vertex is the native comparator; the other three cells are controls, not the published AC-FW algorithm. No curvature or reference quantities are supplied to their step rules.

There are 280 runs with the previous outer cap and tolerance. All 70 AC/vertex runs match the earlier native comparator's objective, certificate, and oracle counts exactly; no new independent instances are added. Table 4 uses separate budgets for function and gradient calls. Each value is the last fully available certificate before that oracle budget, reconstructed from the complete trace before sampling. It is not an interpolation between saved trace rows. Other oracle types are

Table 3: 350-run extension, selected families. Medians include capped runs. The cap is 1000 outer iterations, including AC-FW rejections; the compulsory AC-FW initialization is included in all oracle costs. Success means $U \leq 10^{-5}$.

Method	Successes	Outer iterations	U	f calls	∇f calls
active					
Ordinary, $M = U_0/D^2$	1/10	1000.0	2.43e-04	2002.0	1001.0
Ordinary, $M = 1$	0/10	1000.0	2.23e-04	1998.5	1001.0
Certificate scaling	1/10	1000.0	1.99e-04	2014.0	1001.0
AC-FW, $\delta = 1$	10/10	16.5	7.93e-06	18.5	18.5
Armijo	10/10	19.0	6.55e-06	141.0	20.0
active_scaled					
Ordinary, $M = U_0/D^2$	0/10	1000.0	1.87e-03	2002.0	1001.0
Ordinary, $M = 1$	0/10	1000.0	1.77e-03	2002.0	1001.0
Certificate scaling	0/10	1000.0	1.47e-03	2014.0	1001.0
AC-FW, $\delta = 1$	0/10	1000.0	2.67e-03	1002.0	1002.0
Armijo	1/10	1000.0	4.78e-04	9758.0	1001.0
bike_poisson					
Ordinary, $M = U_0/D^2$	6/10	500.0	9.87e-06	1002.5	501.0
Ordinary, $M = 1$	8/10	893.5	9.85e-06	1789.0	894.5
Certificate scaling	8/10	419.0	9.41e-06	860.0	420.0
AC-FW, $\delta = 1$	7/10	637.0	9.53e-06	639.0	639.0
Armijo	10/10	310.0	9.93e-06	4399.0	311.0

unrestricted. All nonterminal runs can reach 1000 function calls before hitting the outer cap, so this primary budget comparison is not censored by that cap.

On the active family, moving to the LMO vertex changes the number of certified runs at 1000 function calls from 1 to 10 for AC and from 0 to 10 for BT. Thus the earlier 10/10 versus 0–1/10 comparison did not isolate the subsequent curvature update. The median paired certificate ratio BT/AC from the vertex is 1.049, with descriptive percentile-bootstrap interval $[0.935, 1.257]$. Vertex/original ratios are 0.0351 for AC and 0.0150 for BT. These describe the ten matched instances at a fixed budget, not speedup factors or a population-level ranking; completed runs stop near the tolerance. All 84 comparisons, including other oracle budgets and families, are recorded with 10000 bootstrap resamples and seed 20261015, without a multiple-comparison correction.

On Bike Sharing at 1000 function calls, AC certifies 6/10 starts from the original point and 7/10 from the vertex; BT certifies 2/10 and 5/10. At 1000 gradients BT/original certifies 6/10, like AC/original. The oracle unit changes the comparison. The scaled active family remains unsolved by all four cells within the reported budgets. Figure 2 shows the function-budget profiles.

A geometric explanation on the planted instances. Let $w = \operatorname{sign}(x^*)$, zero off its support, and $\sigma(x) = R - w^\top x \geq 0$. The optimal face is $F = \{x \in Q : \sigma(x) = 0\}$. Every FW update satisfies exactly

$$\sigma(x + \alpha(s - x)) = (1 - \alpha)\sigma(x) + \alpha\sigma(s).$$

If selected vertices lie in F and all steps are strictly below 1, a point outside F retains positive slack at every finite iteration. A full initial move to a vertex in F removes this slack immediately. For the planted normal v, $|v_i| \leq 0.6$ off the support, so

$$f(x) \geq \lambda(R - v^\top x) \geq 0.4\lambda\sigma(x).$$

The last inequality uses $\|x_{\text{off}}\|_1 \leq \sigma(x)$. This is an application of the familiar face geometry of vanilla FW [10], not a new acceleration mechanism.

Table 4: Controlled starting-point/update-rule ablation. Each cell has the same preprocessing cost and first trial curvature. Successes and median certificates at separate budgets of 1000 function or gradient calls, over ten seeds. Other oracle types are unrestricted; the outer cap remains 1000.

		1000 function calls		1000 gradient calls	
Rule	Start	Successes	Median U	Successes	Median U
active					
AC	original	1/10	2.06e-04	1/10	2.06e-04
AC	vertex	10/10	7.93e-06	10/10	7.93e-06
BT	original	0/10	5.18e-04	0/10	2.61e-04
BT	vertex	10/10	8.44e-06	10/10	8.44e-06
active_scaled					
AC	original	0/10	2.00e-03	0/10	2.00e-03
AC	vertex	0/10	2.68e-03	0/10	2.68e-03
BT	original	0/10	3.72e-03	0/10	1.86e-03
BT	vertex	0/10	3.60e-03	0/10	1.84e-03
bike_poisson					
AC	original	6/10	9.72e-06	6/10	9.72e-06
AC	vertex	7/10	9.53e-06	7/10	9.53e-06
BT	original	2/10	6.72e-05	6/10	9.94e-06
BT	vertex	5/10	2.20e-05	5/10	1.11e-05

Post-run diagnostics replay all 120 controls in the three active families, using the planted support only after solving. The slack identity and lower bound hold throughout. On the active family the initial LMO vertex lies in F for all ten seeds, and both vertex-start cells stay in F. BT/original takes no full step and retains positive final slack in all ten runs; AC/original takes one full step in its sole successful run. At the outer cap, BT/original's smallest final slack among these ten runs is about $9.98 \cdot 10^{-4}$. Since $\lambda = 0.15$, the lower bound above already exceeds the target 10^{-5} in every run; failure here is not merely a loose stopping certificate. In the high-dimensional family the initial vertex is in F for eight seeds, and the corresponding vertex-start runs stay there. In the scaled family only one initial vertex is in F, and no vertex-start run remains in F throughout. This explains the observed initialization sensitivity on these constructed examples without a claim for general applications.

9.7 Away-step comparison from the original point

The initialization study motivates a direction-level control. We compare FW and AFW, each with ordinary backtracking (BT) or the explicit known-H quadratic step (Q), on the same 70 instances. All four methods start at the original vertex, without a compulsory initial move. BT starts at estimate 1, halves before each search, and doubles on rejection. Q uses $H_0 + H_1 U_k$. AFW chooses the stronger FW/away direction and respects its active-weight cap. The certificate uses the full FW gap even on away steps. This is an implementation of established algorithms, not a newly proposed AFW direction rule.

The 280 runs include 140 exact repeats of the earlier two FW baselines, verified against their saved objectives, certificates and oracle counts. The 140 AFW runs add method comparisons on existing instances, rather than new independent tasks. Table 5 distinguishes the outer cap of 1000 updates from a fixed budget of 1000 function calls. The budget values use complete trajectory prefixes. Function, gradient and full-LMO calls are reported separately; each method uses $N + 1$ gradients and full LMOs. AFW additionally scans its active set once per update, and records the number of atom scores. Maintaining and reconstructing active weights also costs work, so oracle

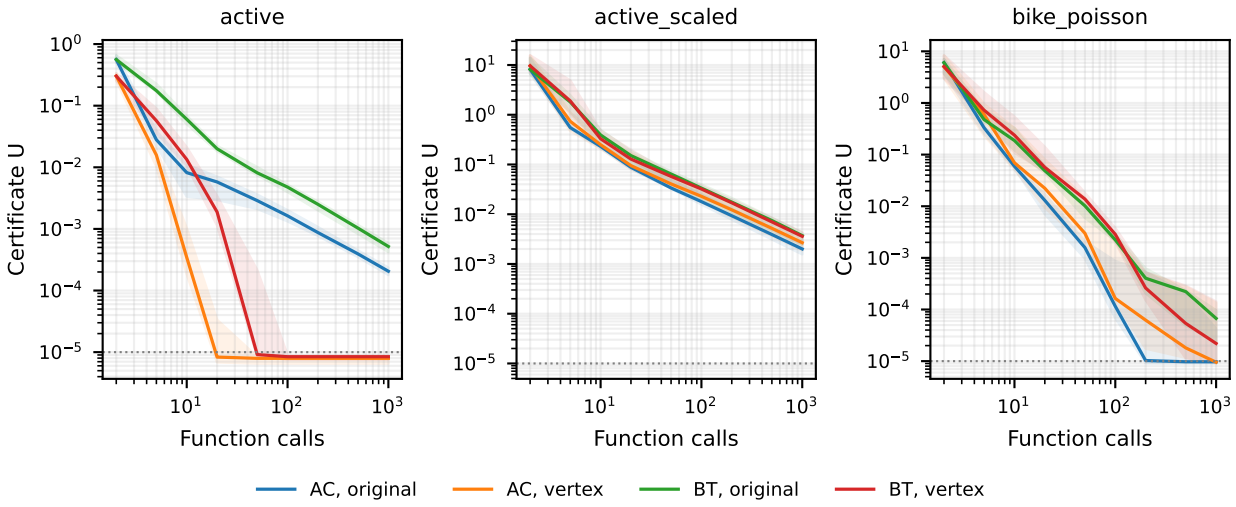


Figure 2: Controlled ablation: median certificate and interquartile band at the displayed function-call budgets. Every prefix includes common preprocessing. Runs stop at $U \leq 10^{-5}$ or 1000 outer trials; completed certificates are carried forward. The displayed function budgets are reached before the outer cap in nonterminal runs. Lines between displayed budgets are for visualization; the CSV contains the actual prefix values.

counts alone do not establish a wall-clock advantage.

AFW-BT reaches $U \leq 10^{-5}$ in all 70 runs within the outer cap. On the scaled active family, it certifies 10/10 instances, using medians 56.5 function calls and 28.5 gradients. FW-BT certifies none within the cap. On the breast-cancer control, AFW-BT certifies 10/10 with 103.5 median function calls, versus 0/10 for FW-BT. On Bike Sharing AFW-BT certifies 10/10 by the end, but only 6/10 at 1000 function calls; the corresponding FW-BT count at that budget is 1/10. End-of-run medians include capped runs and must be read with these successes.

AFW-Q still certifies no run of either real regression within 1000 updates. Thus active-set correction alone does not make a conservative analytic step efficient on these data. On the interior family, the FW/AFW differences are modest. Figure 3 shows the fixed function-budget profiles; it does not establish a ranking over all projection-free methods or untested instances.

Every step of the 70 AFW-BT trajectories is checked before trace sampling: accepted-curvature envelope, dyadic trial identity, local decrease, phase lag budget, drop count, and total function-call bound. The certificate bridge is checked using the independent reference interval and the analytically valid ridge lower bound on strong convexity. No positive weights are pruned heuristically in these runs, and no unexpected numerical atom deletions occur. A separate interval test, where the geometric width is explicit, checks (16) and the strongly convex phase bound. A geometric constant is not fitted to the benchmark trajectories, and these finite checks do not independently validate the general theorem.

9.8 Radius and objective-unit controls

The final control varies the radius while keeping each real-data objective fixed. The logistic radii are $1, 2, 4$ and the Poisson radii are $0.75, 1.5, 3$. Each radius has a newly computed constrained reference interval; the Poisson curvature bounds are recomputed on its new domain. Five matched seeds and objective scales $a \in \{10^{-3}, 1, 10^{3}\}$ give 360 runs of four methods: FW/AFW with either

Table 5: FW versus AFW from the same original point, ten matched seeds. Final medians include capped runs; the cap is 1000 updates. The last columns use a separate budget of 1000 function calls.

Method	Final successes	f calls	∇f calls	Successes at 1000 f	Median U
active_scaled					
FW_BT	0/10	2002.0	1001.0	0/10	3.28e-03
AFW_BT	10/10	56.5	28.5	10/10	7.94e-06
FW_Q	0/10	1001.0	1001.0	0/10	4.26e-03
AFW_Q	10/10	160.5	160.5	10/10	8.67e-06
breast_cancer					
FW_BT	0/10	1997.0	1001.0	0/10	3.11e-04
AFW_BT	10/10	103.5	53.5	10/10	9.52e-06
FW_Q	0/10	1001.0	1001.0	0/10	8.67e-03
AFW_Q	0/10	1001.0	1001.0	0/10	1.24e-03
bike_poisson					
FW_BT	8/10	1789.0	894.5	1/10	3.01e-05
AFW_BT	10/10	877.5	439.0	6/10	9.88e-06
FW_Q	0/10	1001.0	1001.0	0/10	1.04e-01
AFW_Q	0/10	1001.0	1001.0	0/10	1.00e-01

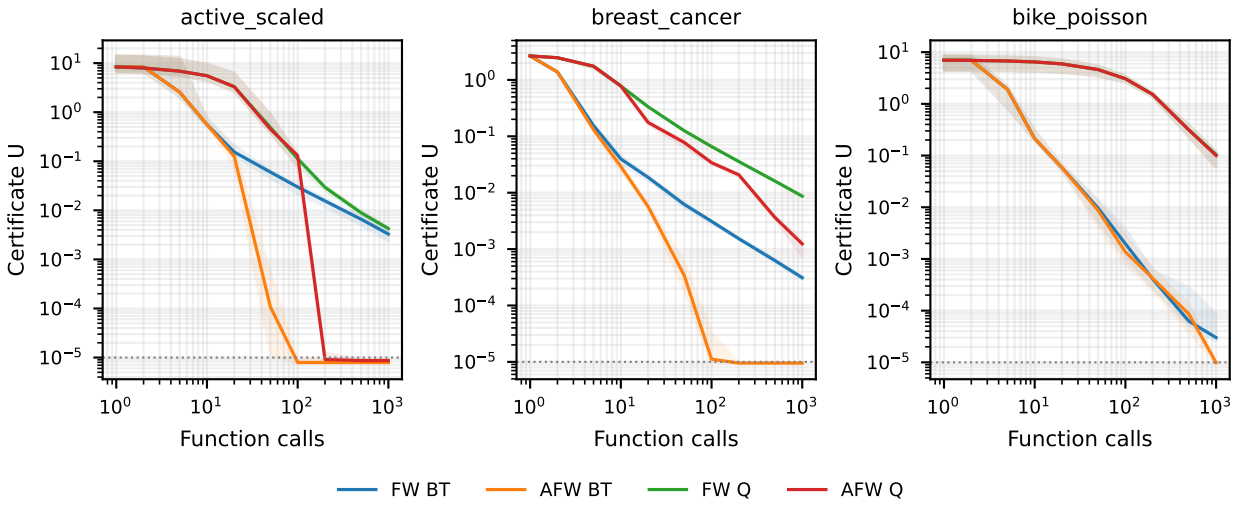


Figure 3: FW versus AFW from a common original start. Median certificate and interquartile band at displayed function-call budgets. Completed runs are carried forward. Every nonterminal run reaches the displayed budgets before the outer cap. Lines between budgets are for display; the CSV stores actual available certificates. Q methods receive analytic curvature constants; BT methods do not. Active-set work is additional.

$M = 1$ or $M = U_0/D^2$. The target for af is $a10^{-5}$, so all conditions use the same accuracy in original units. No reference or curvature constants are solver inputs, and certificate initialization reuses the first paid function, gradient, and LMO evaluations. The 30 unscaled default-radius baselines agree exactly with their earlier records.

Proposition 16 (Objective units in halving backtracking)**.** *In exact arithmetic, replace f by $\widetilde{f} = af + b$, $a > 0$, and ε by $a\varepsilon$. For FW or AFW, scale M by a and use identical scale-invariant tie breaking. Then all iterates, accepted steps, trial counts, and stopping indices are unchanged. In particular, $M = U_0/D^2$ makes this scaling automatic. Keeping M fixed instead corresponds to initialization M/a on the original objective.*

Proof. Every tangent lower bound transforms as $\widetilde{\ell} = a\ell + b$, and $\widetilde{G} = aG$, $\widetilde{U} = aU$. Positive scaling preserves LMO and away-atom comparisons. Inductively $\widetilde{L} = aL$, so the ratio of directional gap to $L\|d\|^2$, the feasible cap, and the quadratic acceptance comparison are unchanged. Halving and doubling commute with positive scaling. The stopping test scales identically. □

This elementary property is an initialization control, not a new convergence result. The curvature parameters transform as $\widetilde{H}_0 = aH_0$, $\widetilde{H}_1 = H_1$; strong convexity scales as $\widetilde{\mu} = a\mu$.

Both AFW initializations certify all 90 radius/seed/scale conditions within 1000 updates. FW certifies 30/90 with fixed M and 33/90 with certificate initialization. These counts include repeated scalings of the same problems and are not independent replications. At scale one, the certificate-initialized AFW function-call median increases from 73 to 388 as the logistic radius goes from 1 to 4; all five seeds still finish. At Poisson radius 3, that median is 1292 and only 1/5 seeds finish within 1000 function calls. Thus success within the update cap is distinct from success at a fixed function budget.

Objective units can affect more than the final call count. For logistic radius 1, FW with fixed $M = 1$ certifies 0/5 at $a = 10^{-3}$ but 5/5 at $a = 1$ and 10^3. For both certificate-initialized methods, all 60 matched scale comparisons preserve final function/gradient/LMO counts and stopping outcomes. The maximum relative change in U/a across the recorded oracle-budget prefixes is below $3.2 \cdot 10^{-11}$, using denominator $\max\{10^{-5}, U_{a=1}\}$. This compares recorded certificates and counters, not bitwise equality of complete iterates; finite precision can change branch decisions in general.

All logistic references are on the constraint boundary. The Poisson reference has ℓ_1 norm approximately 0.692 and is interior at all three radii. Thus this control does not test an active Poisson constraint. No new dataset or predictive-performance claim is added. Changing the radius changes both the domain and its original starting vertex. Certificate initialization is not uniformly faster than the fixed initialization. Additional AFW active-set work remains separately counted, and no wall-clock ranking follows.

9.9 A Poisson constraint-activity transition

The preceding Poisson radii all leave the reference interior. We now fix that objective and vary R/R_* over the seven predefined values $0.5, 0.9, 0.99, 1, 1.01, 1.1, 2$. Here $R_* \simeq 0.691643204469$ is the ℓ_1 norm of an independently computed stationary point. Ridge strong convexity gives the residual estimate

$$\left|\|x_{\mathrm{ref}}\|_1 - \|x^*_{\mathrm{unc}}\|_1\right| \le \sqrt{d}\,\|\nabla f(x_{\mathrm{ref}})\|\,/\mu.$$

The numerical right-hand side is about $4.1 \cdot 10^{-15}$. These are floating-point residual calculations, not interval arithmetic. The radius equal to the numerical estimate is marked transitional; the other regimes are separated from the estimate by much larger margins. The three smaller ratios

Table 6: Radius controls at objective scale $a = 1$, five matched seeds. Each entry gives final successes (out of five); median function calls. Capped failures are included. The cap is 1000 updates, not 1000 function calls. Subscript 1: $M = 1$; subscript U: $M = U_0/D^2$.

Radius	FW_1	FW_U	AFW_1	AFW_U
breast_cancer				
$R = 1$	5; 205	5; 161	5; 83	5; 73
$R = 2$	0; 1997	0; 1999	5; 105	5; 101
$R = 4$	0; 1998	0; 2000	5; 342	5; 388
bike_poisson				
$R = 0.75$	0; 2002	0; 2003	5; 829	5; 906
$R = 1.5$	4; 1804	4; 423	5; 791	5; 635
$R = 3$	3; 1920	2; 1999	5; 1085	5; 1292

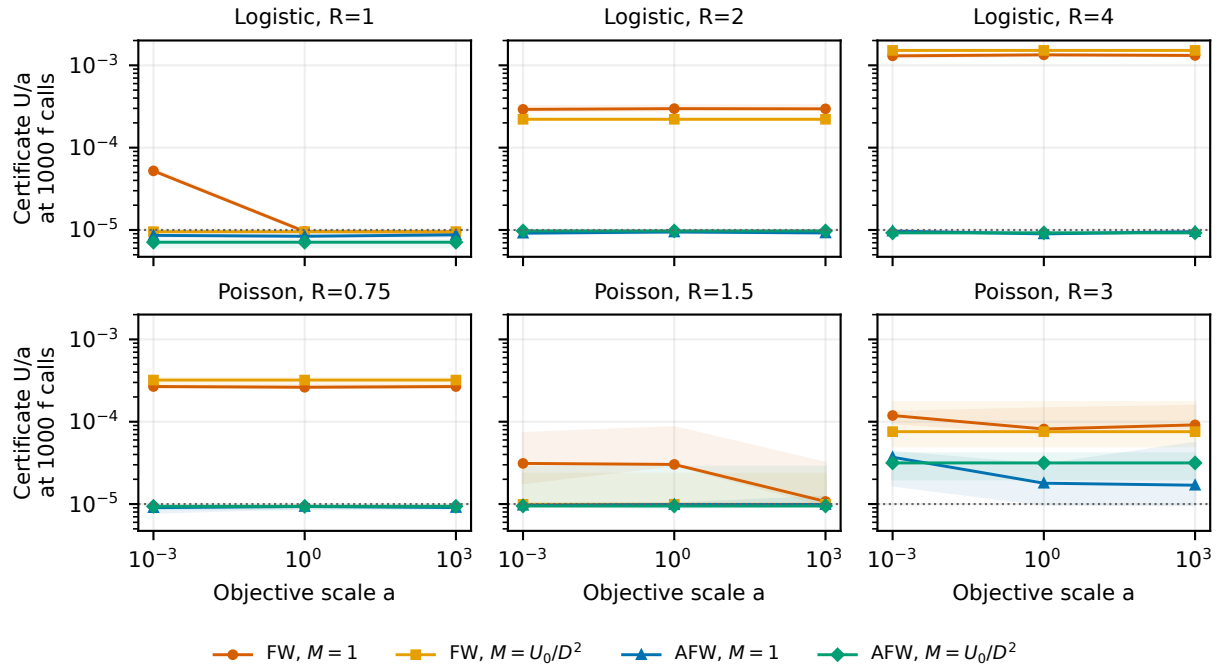


Figure 4: Radius and objective-unit controls. Median and interquartile band of the certificate in original units, U/a, after at most 1000 function calls, five matched seeds. The tolerance is 10^{-5} in these units. Completed runs are carried forward; nonterminal runs reach the displayed budget before the update cap. Lines only connect the three tested scales. Scaled versions of a problem are paired controls.

give active references with positive estimated KKT multipliers; the three larger ones give interior references. References and activity information are unavailable to the algorithms.

There are 280 raw trajectories of the same four FW/AFW backtracking variants, to $U \le 10^{-7}$ or 2000 updates. Their nested prefixes record first attainment of $10^{-3}, 10^{-5}, 10^{-7}$ and separate oracle budgets. Ten seed labels produce only seven distinct initial vertices: seeds 6, 7, and 8 repeat seeds 1, 0, and 2, respectively. All stored non-timing records for those 84 repeated trajectories agree exactly. The primary table and figure retain one representative per vertex, giving 196 distinct method/radius/start trajectories. These share one dataset and one objective; they are not independent dataset replications. Earlier seed-weighted tables retain their original protocol, and should also be read with this multiplicity in mind.

Both AFW variants attain 10^{-7} on all 49 distinct radius/start conditions. FW with $M = 1$ attains it on 4/49, and FW with $M = U_0/D^2$ on 5/49. In the transitional region neither FW variant reaches 10^{-7} within the cap. At $R/R_* = 2$, certificate initialization allows FW to finish on 4/7 starts, with median final cost 870 function calls including capped failures; AFW finishes on 7/7, with median 1946 for that initialization. Capped medians alone therefore do not define a ranking. The certificate initialization is also slower than $M = 1$ for AFW at several radii. Active constraints, starting vertices, solution support, and conditioning all change with radius; this is not an isolation of one geometric constant or a universal ranking.

Every source of all 280 raw trajectories is checked against the local gap envelope in Lemma 12, using $\delta \le f(x) - f^*_{\mathrm{ref,lower}}$. The new threshold reduces the analysis phase count before rounding by about 15–34 compared with the previous distance-based threshold on these cases. It changes the proof, not the algorithm or measured cost. No geometric constant κ is fitted. Estimated reference supports use threshold 10^{-7}; their final face-slack diagnostics do not certify exact finite face identification. Active-set work is additional to the function, gradient, and full-LMO counts.

Table 7: Poisson transition sweep, 7 distinct starting vertices per radius. Entries are successes at 10^{-7} (out of 7); median final function calls, including failures capped at 2000 updates. Subscripts give $M = 1$ or $M = U_0/D^2$.

R/R_*	FW_1	FW_U	AFW_1	AFW_U
0.5	2; 4002	1; 4003	7; 1399	7; 1043
0.9	0; 4002	0; 4003	7; 1036	7; 1331
0.99	0; 4001	0; 4003	7; 1150	7; 1585
1	0; 4002	0; 4003	7; 1387	7; 1842
1.01	0; 4001	0; 4003	7; 1396	7; 1862
1.1	0; 4002	0; 4003	7; 1360	7; 1730
2	2; 4001	4; 870	7; 1697	7; 1946

9.10 Two additional count datasets and a limit of the certificate

We add Seoul Bike Sharing Demand [18] and Metro Interstate Traffic Volume [19]. The official files, their SHA-256 values, and CC BY 4.0 attribution are retained. A stable chronological sort followed by the first 80% of rows gives 7008 Seoul observations with 15 model coordinates and 38563 Metro observations with 12 coordinates. Repeated source timestamps retain their original weights. Fixed feature choices include weather, cyclic time features, holiday, and Seoul's functioning indicator; other textual categories are omitted. Each non-intercept feature becomes $\tanh((v - \bar{v}_{\mathrm{train}})/s_{\mathrm{train}})$ using training statistics only. This bounded-feature protocol differs from the earlier Bike Sharing standardization. Predictive accuracy is not evaluated.

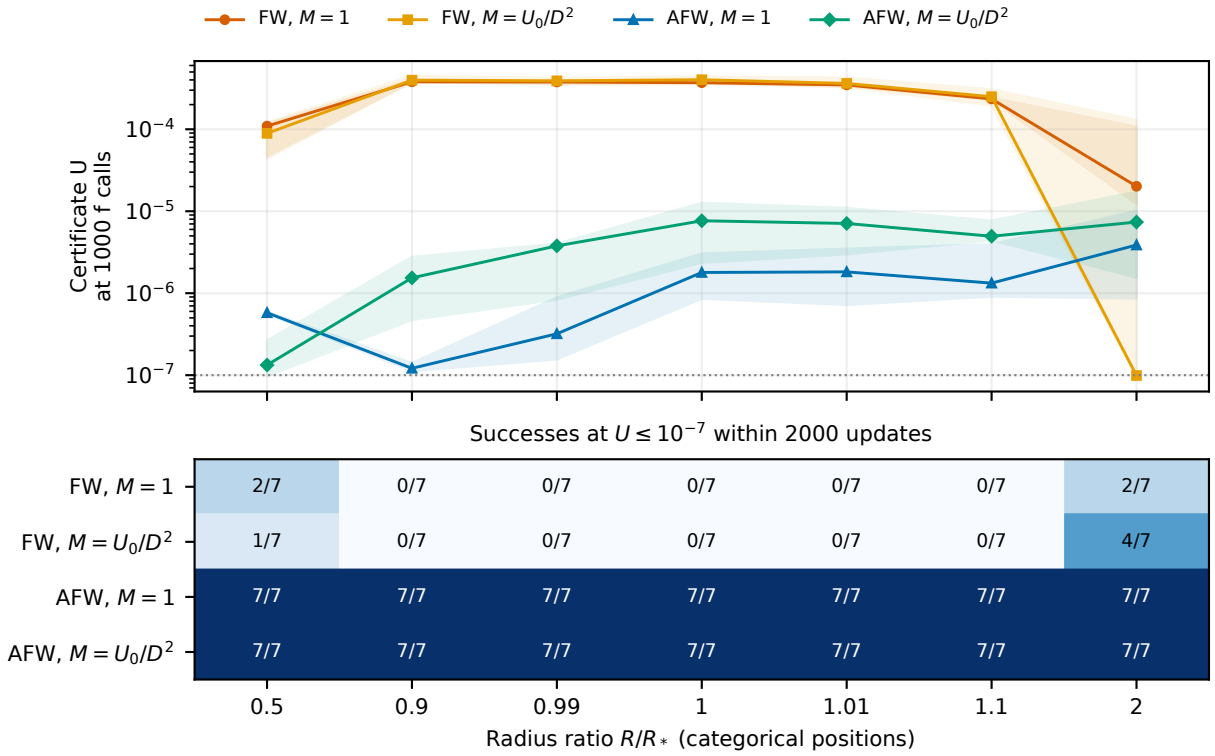


Figure 5: Poisson activity sweep with equal weights on seven distinct starts per radius. Top: median and interquartile certificate at 1000 function calls; completed runs are carried forward, and nonterminal runs reach this budget. Bottom: successes at 10^{-7} within 2000 updates. Radius ratios occupy categorical positions, so their spacing does not represent Euclidean distance. This is one dataset.

The normalized Poisson deviance and ridge coefficient $\mu = 0.01$ are as before; the intercept is also constrained and regularized. Since $\|a_i\|_\infty = 1$ and the normalized response t_i has mean one, valid radius-R bounds are

$$H_1 = \max_i \|a_i\|^2, \quad H_0 = \mu + H_1(1+R), \quad L_Q = \mu + e^R \lambda_{\max}(A^\top A/m).$$

The Hessian norm is at most $\mu + H_1 \operatorname{mean}(e^{a_i^\top x})$; comparing f_Q^* with $f(0)$ yields $\operatorname{mean}(e^{a_i^\top x}) \le \delta + 1 + R$. Thus the condition holds relative to the constrained minimum. These are upper bounds, not minimal curvature parameters, and positive H_1 is not claimed necessary on a compact domain.

Offline damped Newton solves estimate $R_* := \|x^*_{\text{unc}}\|_1$. The values are approximately 4.575821 (Seoul) and 1.431745 (Metro). We compare $R/R_* = 1/2$ and 2, with independently recomputed active and interior reference intervals of width below $4.3 \cdot 10^{-8}$. All 30 Seoul or 24 Metro signed coordinate vertices are used once per radius. Six existing variants give 648 trajectories: the four BT variants and FW/AFW known-H short steps. The cap is 1000 updates, the tolerance 10^{-5}. References, activity and R_* are offline information, not solver inputs. Starts and radii within each dataset share data.

Both AFW-BT variants finish all 30 active Seoul and all 24 active Metro starts. Their function-call medians are 424 versus 399 and 99 versus 80.5 for $M = 1$ versus $M = U_0/D^2$, respectively. The interior controls are unfavorable: neither AFW-BT variant finishes any Seoul start, and only 6/24 and 7/24 finish on Metro. AFW-Q finishes all active Metro starts with median 423 function/gradient calls, but no start in the other three cases. These results limit any practical-success claim based on the earlier Bike Sharing controls. All full trajectories pass descent, feasibility, trial-count, active-removal, and local-gap checks before sampling.

The independent reference interval distinguishes primal error from the online certificate. At the interior Metro radius, every BT run has post-run upper error $e_{\text{ref}} = \max\{0, f(x) - f^*_{\text{ref,lower}}\} \le 10^{-5}$. Yet FW has no online U successes and AFW has only 6 and 7. On interior Seoul, 12/30 and 18/30 AFW runs have $e_{\text{ref}} \le 10^{-5}$, although neither variant stops. Noncompletion can therefore reflect both remaining primal error and a conservative observable certificate. References do not replace actual stopping outcomes. Tighter dual certificates merit investigation with their extra linear oracles properly counted. Active-set work is additional, and no wall-clock ranking follows from these counts.

Table 8: Additional count datasets, every signed coordinate vertex. Entries: successes; median final function calls including failures capped at 1000 updates. Seoul has 30 starts per radius, Metro 24. F/A: FW/AFW; subscripts $1, U$: initial $M = 1, U_0/D^2$; Q: known-H short step.

R/R_*	F_1	F_U	A_1	A_U	F_Q	A_Q
seoul						
0.5	1; 2001	1; 2003	30; 424	30; 399	0; 1001	0; 1001
2	0; 2001	0; 1996	0; 2001	0; 1995.5	0; 1001	0; 1001
metro						
0.5	3; 2001	3; 2003	24; 99	24; 80.5	1; 1001	24; 423
2	0; 2000	0; 2002	6; 2000	7; 2001	0; 1001	0; 1001

9.11 Auto-conditioned comparison on the additional count datasets

We add 108 AC-FW runs to the same 108 Seoul/Metro cases, preserving the 648 earlier records. Algorithms 1–2 and Eq. (6) with $\delta = 1$ of [16] are implemented independently, with the mandatory

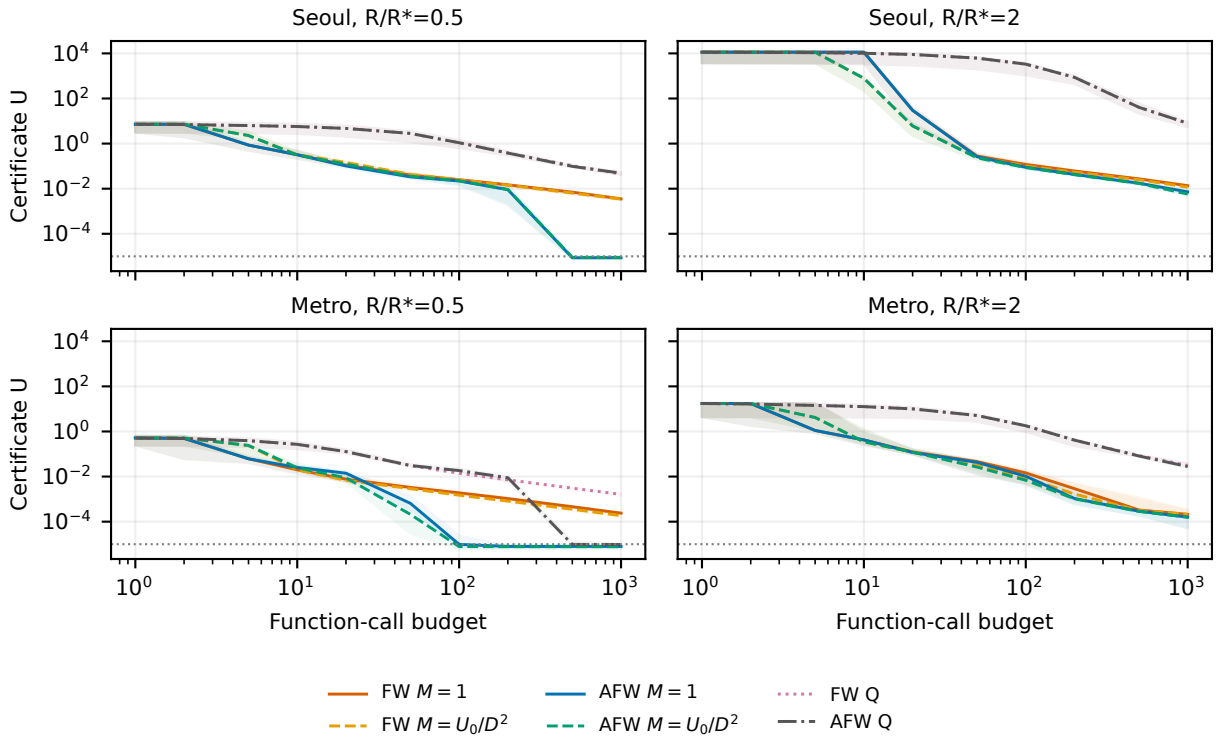


Figure 6: Additional real count datasets. Median certificate and interquartile band over every signed coordinate vertex, at separate function-call budgets. Completed runs are carried forward; nonterminal runs reach the displayed budgets. BT has no curvature inputs; Q receives analytic H_0, H_1. Radii and starts within a dataset share training data.

initial vertex move and both initial source evaluations charged. Rejected trials consume the 1000-outer-trial cap. Actual counts are $N_f = N_{\mathrm{LMO}} = N + 2$ and $N_g = N_{\mathrm{accepted}} + 2$. Parameter-hidden tests on both datasets instrument actual calls.

The common budget of 1000 function calls is reached by every nonterminal run before its cap. AC-FW certifies 13/24 active Metro starts, with median endpoint function count 385, and no start in the other three dataset/radius conditions. FW-BT has 2/24 and 3/24 active Metro successes at this budget, while both AFW-BT variants have 24/24. On active Seoul, AFW-BT has 30/30 and AC-FW has none. All seven scalar-certificate variants remain unsuccessful on both interior domains at 1000 function calls. These are oracle-budget observations, not a uniform performance or runtime ranking.

The exponential objective is not globally gradient-Lipschitz on $\mathbb{R}^d$, so this remains an empirical use on compact Q rather than a verbatim application of the cited global theorem. The new archive contains 3240 budget prefixes and full-source audits. The focused article joins these records with the six preserved methods in its generated tables and vector figure.

9.12 Paid causal certificate controls

We repeat the 432 BT trajectories from Section 9.10, using the same 108 dataset/radius/vertex cases and four methods. All original terminal values, certificates, update counts and oracle counts match the earlier archive exactly. The five controls are scalar, disjoint blocks of $K = 8, 32, 128$, and known $\mu = 0.01$, as defined in Section 7. Block sizes are fixed before observing the results. These give 2160 dependent causal stopping profiles, not 2160 independent optimization runs. No new data or starting points are introduced.

The monitors see only paid source values and gradients; neither the line search nor initialization changes. We record the first causal prefix with certificate at most 10^{-5}, or the original endpoint. Twenty native online-stop checks on a synthetic interior instance verify the prefix construction and actual oracle costs for every method/profile pair. Each profile's isolated LMO cost includes its own block queries and the direction queries, excluding the other monitors. An affordable budget prefix must pay the entire scheduled verification at its source. No incomplete block is queried, and references are not monitor inputs. The original trajectory cap is retained.

All three block sizes fail to produce any earlier stop: each retains 129/432 successes. At the common original endpoint, $K = 8$ improves the certificate on 61 paths and $K = 32$ on 22, but the improvements do not cross the tolerance. Blocks increase total LMO cost on 432, 425, and 380 paths, respectively. This negative control does not establish failure of other weights, block sizes, or exact Fenchel dual evaluation.

The known-μ control gives 226/432 successes and 110 earlier prefixes without additional LMOs. On interior Metro all four methods certify 24/24 starts. FW's function-call medians become 746.5 and 724.5, versus censored scalar medians 2000 and 2002; AFW's become 481.5 and 458, versus 2000 and 2001. At the separate budget of 1000 function calls, FW has 18/24 and 15/24 successes, and both AFW variants have 24/24. Interior Seoul remains difficult: AFW has 4/30 and 10/30 successes at the full cap, but none at 1000 function calls; FW has none. Active-regime success counts are unchanged.

Thus a standard certificate using analytical model information resolves much of the observed Metro stopping lag, while the tested first-order block controls do not. The known-μ result is not evidence of adaptation to unknown strong convexity or a new rate. The original scalar-certificate bounds remain the main analytical claim. Full-path audits check lower-bound validity, certificate monotonicity, dominance over scalar, and the exact extra-LMO schedule before sampling. Active-set scans remain additional work; no wall-clock comparison is made.

Table 9: Interior causal stopping controls. Each cell is the certified count and median function calls up to its first stop or the original cap; censored cases are included. Seoul has 30 starts per method, Metro 24. F/A: FW/AFW; subscripts $1, U$: initial $M = 1, U_0/D^2$. Blocks use one additional full LMO per completed block; the known-μ column receives extra problem information.

Data	Method	Scalar	$K = 8$	$K = 32$	$K = 128$	Known μ
Seoul	F_1	0; 2001	0; 2001	0; 2001	0; 2001	0; 2001
Seoul	F_U	0; 1996	0; 1996	0; 1996	0; 1996	0; 1996
Seoul	A_1	0; 2001	0; 2001	0; 2001	0; 2001	4; 2001
Seoul	A_U	0; 1995.5	0; 1995.5	0; 1995.5	0; 1995.5	10; 1995
Metro	F_1	0; 2000	0; 2000	0; 2000	0; 2000	24; 746.5
Metro	F_U	0; 2002	0; 2002	0; 2002	0; 2002	24; 724.5
Metro	A_1	6; 2000	6; 2000	6; 2000	6; 2000	24; 481.5
Metro	A_U	7; 2001	7; 2001	7; 2001	7; 2001	24; 458

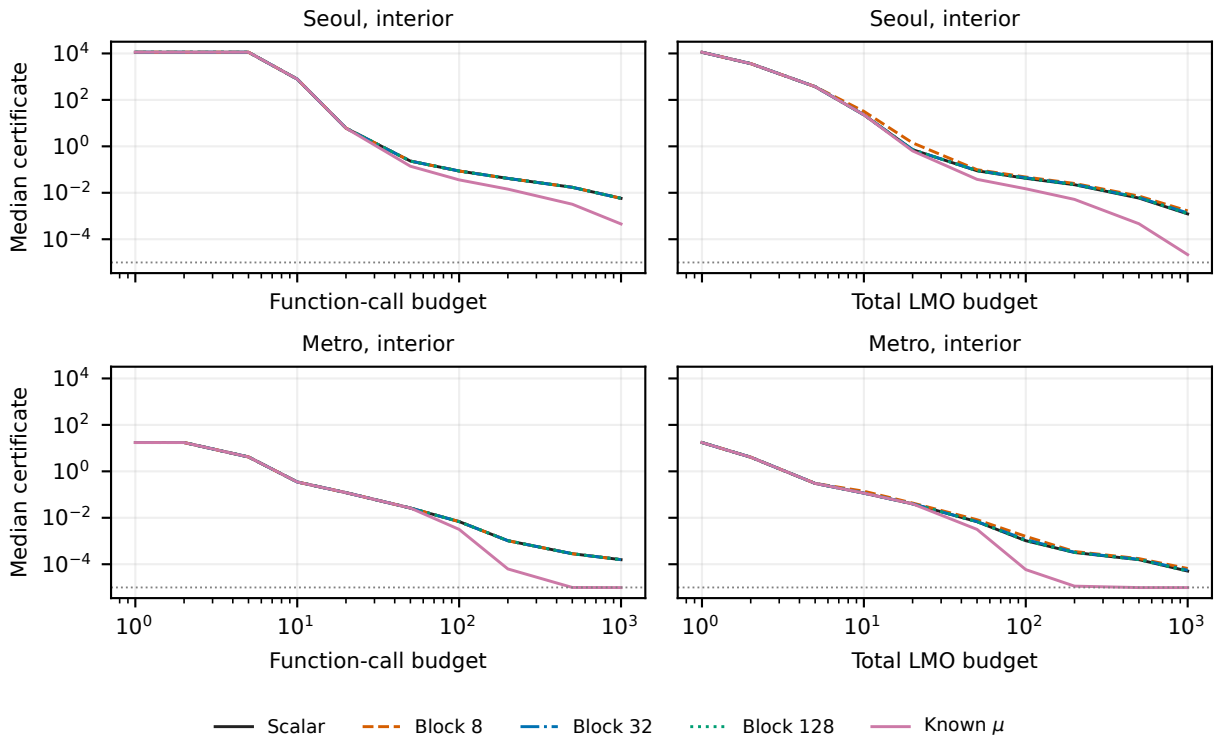


Figure 7: Causal controls on interior AFW-BT paths with initial $M = U_0/D^2$. Median certificates over all 30 Seoul or 24 Metro vertices at separate function and total LMO budgets; total LMOs include the profile's block queries. Completed prefixes are carried forward. Known μ receives analytical ridge information; blocks do not. All methods and both initializations appear in Table 9 and the archived report.

10 Discussion

The main candidate result is a two-phase certificate analysis of ordinary quadratic backtracking without smoothness inputs, counting both LMO and all function trials. Overestimated curvature requires only an additional phase-controlled number of steps. Certificate normalization is optional: the order alone does not justify it as a new algorithmic contribution. The LMO guarantee also holds for the explicit rules and exact line search. The empirical results support reporting unfavorable regimes, controlling initialization, and separating oracle costs. On a planted active family, the compulsory vertex move largely explains the earlier apparent advantage of AC-FW over the chosen backtracking baselines. Away steps provide a standard way to remove obstructing atoms without a special initial move. Their established geometric progress inequality also yields our phase bound under an additional geometric error bound. The certificate conversion itself needs no strong convexity: feasible descent provides a local gap envelope. Only the geometric error bound is required separately in the logarithmic phase theorem. The known linear convergence of AFW is not a novelty claim; the candidate specialization concerns the separated curvature dependence, this stopping certificate, and total trial cost. The radius/units control also shows that fixed numerical curvature initialization can affect outcomes on equivalent objectives; matching the units of M is necessary for an objective-scale comparison. On the additional count datasets, success in active regimes does not extend uniformly to larger interior domains. Independent reference intervals expose a gap between primal accuracy and observable stopping. The tested block tangent bounds do not improve stopping and add LMO cost. A standard known-μ control improves Metro stopping without extra LMOs, showing the role of available model information. Its residual failures on Seoul limit a general claim. Neither control adds a new duality or convergence result.

Open work includes a complete comparison with certificate/restart and relative-smoothness analyses, an adaptive sliding baseline, independent validation of the proof, more natural positive-H_1 applications, and pairwise extensions. Noisy directions and finite-precision robustness require additional theory. Convexity of the feasible region ensures that segment updates are feasible; a transfer to an ℓ_0 constraint does not follow from the present argument.

A What direct substitutions into existing bounds establish

This comparison concerns the stated bounds below, rather than every consequence of the cited papers. It does not establish bibliographic priority.

Pedregosa et al. [11], Theorem 3, give a convex primal–dual bound in terms of averaged accepted smoothness estimates. Thus adaptive curvature and last-iterate dual guarantees are already present there. One cannot identify that dual value with the maximum of scalar tangent lower bounds used here: their dual variable averages gradients. This distinction alone does not prove a new rate; a transfer argument along the algorithm's trajectory could still exist.

Here is an elementary illustration of the certificate distinction. For $f(x) = x^2/2$ on $[-1, 1]$, the scalar tangent lower bound is $a(x) = -x^2/2 - |x|$, whereas the Fenchel dual value is $\psi(u) = -u^2/2 - |u|$. For the history $(-1, 1)$,

$$\max_i a(x_i) = -3/2 < \psi((x_0 + x_1)/2) = 0.$$

For the history $(-1, -1/2)$,

$$\max_i a(x_i) = -5/8 > \psi(-3/4) = -33/32.$$

Thus the two lower bounds have no universal ordering for arbitrary tangent histories. These histories are not asserted to be iterates of the cited algorithm; neither example contradicts its theorem.

A global Euclidean-smoothness substitution also loses information. Let $\Delta_Q = \max_{x \in Q}(f(x) - f_Q^*)$. Assumption (1) gives a valid bound $L_Q = H_0 + H_1\Delta_Q$ on Q. Applying a conventional $O(L_Q D^2/\varepsilon)$ estimate retains Δ_Q/ε, rather than the logarithmic H_1 term. Replacing Δ_Q by U_0 is not automatically valid: U_0 only bounds the initial suboptimality. Restricting Q to an initial sublevel set changes its LMO. Our local acceptance argument avoids that restriction, followed by a phase-wise count of estimates that remain too large. A finer use of an existing average-curvature bound remains an open priority question.

The relative-smooth framework of Takahashi, Pokutta, and Takeda [20] uses a kernel, a segment scaling condition, and additional assumptions for its accelerated conclusions. Its Euclidean specialization recovers ordinary smoothness bounds, so choosing a Euclidean kernel and $L = L_Q$ alone does not yield the separated H_0, H_1 dependence above. This does not exclude a more suitable kernel or a refined specialization of that analysis.

For adaptive sliding [21], Corollary 3.3 concerns outer iterations with a running maximum of local curvature estimates; Theorem 2.1 separately bounds inner LMO effort. Its restarted Theorem 4.1 additionally assumes local strong convexity and uses local smoothness and strong-convexity parameters in the restart schedule. Our curvature assumption does not imply strong convexity: for example, $f(x_1, x_2) = \cosh(x_1) - 1$ on $[-1, 1]^2$ has $H_0 = H_1 = 1$ and a line of minimizers. Consequently the restarted strong-convexity theorem cannot be invoked from our assumptions alone. A complete comparison must count all inner LMO calls and preserve the stopping certificate; we do not establish superiority over sliding methods.

Wirth, Peña, and Pokutta [2], Definition 2.3(b) and Theorem 3.6, use a uniform extended-curvature constant and a vertex error bound for exact line search or a short step. Their linear and two-regime rates are prior results. Directly substituting a global $L_Q D^2$ retains global curvature; a varying-curvature extension would require further analysis.

Kerdreux, d'Aspremont, and Pokutta [3], Proposition 4.2 and Theorem 5.1, already combine burn-in, computable strong-Wolfe stopping, unknown error-bound parameters, and cumulative LMO counts. The stated bound uses global away curvature. Our phases analyze ordinary backtracking without algorithmic restarts; this difference does not itself establish a novel bound.

Zhao [4], Theorems 4.1–4.2, establishes linear rates for both objective error and FW gap with logarithmically homogeneous barriers. Algorithm 1 uses exact search or a local-Hessian analytic step, rather than our first-order quadratic backtracking. The function assumptions differ. The passage from primal to gap rates is therefore also prior in spirit, and is not claimed as novel here.

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